



DZMS Administrator's Guide

For software version DZMS 4.x

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ABOUT THIS GUIDE

DZS Management System (DZMS) is the network management system that provides service-to-subscriber management for DZS local loop network elements.

DZMS includes a comprehensive suite of applications that deliver fault, configuration, performance, and security management functions as well as operator account management. DZMS Clients, NetHorizon™ and DZMS Web Client, are the applications that operators use to navigate, monitor, and manage objects in their networks. DZMS Web Client is the application that operators can use for the same operations as NetHorizon but for optical and VDSL networks only.

Audience

This guide is written for the system administrator who installs and administers the DZMS system and who sets up and manages operator accounts. As a reader of this guide, you should be familiar with basic networking components such as network devices, cards, physical ports, logical interfaces, and permanent virtual circuits (PVCs). You should also be familiar with network administrative tasks such as managing network components and operator accounts.

Document organization

This guide contains the following information:

Read	To learn about
Chapter 1, Overview, on page 17.	System components and description.
Chapter 2, System Administration, on page 23.	Administering DZMS components.
Chapter 3, Administering Operator Accounts, on page 59.	Administering operator accounts.

Read	To learn about
<i>Chapter 4, DZMS database server administration, on page 81</i>	Administering the DZMS database.
<i>Chapter 5, Monitoring DZMS Logs, on page 93.</i>	Monitoring logs.
<i>Appendix A, How DZMS Manages Network Elements, on page 103.</i>	Details of how DZMS manages resources on the network.
<i>Appendix B, DZMS Legacy Traps and Alarms, on page 119.</i>	How SNMP traps from devices are mapped to DZMS alarms.
<i>Appendix C, Modifying DZMS alarm severities and descriptions, on page 125.</i>	Modifying DZMS alarm severities and descriptions.

Style and notation conventions

The following conventions are used in this document to alert users to information that is instructional, warns of potential damage to system equipment or data, and warns of potential injury or death. Carefully read and follow the instructions included in this document.



Caution: A caution alerts users to conditions or actions that could damage equipment or data.



Note: A note provides important supplemental or amplified information.



Tip: A tip provides additional information that enables users to more readily complete their tasks.



WARNING! A warning alerts users to conditions or actions that could lead to injury or death.



WARNING! A warning with this icon alerts users to conditions or actions that could lead to injury caused by a laser.

Typographical conventions

The following typographical styles are used in this guide to represent specific types of information.

Bold	Used for names of buttons, dialog boxes, icons, menus, profiles when placed in body text, and property pages (or sheets). Also used for commands, options, parameters in body text, and user input in body text.
<code>Fixed</code>	Used in code examples for computer output, file names, path names, and the contents of online files or directories.
Fixed Bold	Used in code examples for text typed by users.
<i>Fixed Bold Italic</i>	Used in code examples for variable text typed by users.
<i>Italic</i>	Used for book titles, chapter titles, file path names, notes in body text requiring special attention, section titles, emphasized terms, and variables.
PLAIN UPPER CASE	Used for environment variables.
Command Syntax	Brackets [] indicate optional syntax. Vertical bar indicates the OR symbol.

Related documentation

Refer to the following publications for additional information:

DZMS Release Notes contains the most current DZMS product information and requirements.

DZMS-HA Platform Installation and Administration Guide describes how to install, monitor, and administrate the DZMS HA Solution.

DZMS Installation Guide describes how to install and configure DZMS.

OSS XML Gateway User's Guide describes how to install, configure, and use the OSS XML gateway application.

OSS XML Gateway Reference Guide describes the attributes associated with configuration and performance requests to the OSS XML Gateway.

NETCONF User Guide describes how to install, configure, and use the OSS NETCONF gateway application.

DZMS Web Client Quick Start Guide describes how to add a device, gives multiple provisioning examples, and discusses provisioning options, service group creation, and the templates based model used in the DZMS Web Client, including procedures.

SLMS GPON & AE Troubleshooting Guide describes how a technician, standing in front of a system, interacts with the hardware components to fix an active alarm (e.g. disconnecting components and viewing LEDs). It also identifies CLI commands that can be used to diagnose and verify when a fault has been fixed. Often times the technician at the equipment sight coordinates

the hardware actions (over a phone) with an administrator that executes CLI commands at a Network Operations Center.

Contacting DZS Quality & Service

All new DZS equipment purchases include one year of HW Warranty and 90 days of Bronze-level Technical Support.

If your product is not within 90 days of the purchase date or you do not have a valid support contract, please contact your local sales representative to get a quote on a support contract.

Customers with a valid support contract or are eligible for 90 days technical support associated with a new product purchase can request technical support by opening a case at:

<https://dzsi.com/support/#TAC>

Customers with a valid support contract have access to technical product documentation, software downloads, knowledge base and consultation on the covered DZS product at the same support portal.

For repair services within the HW Warranty period or under an Extended Warranty support contract, a Return Material Authorization (RMA) must be obtained before sending the equipment for repair. RMA requests can be submitted at:

<https://dzsi.com/support/#RMA>

Technical support

The Technical Assistance Center (TAC) is available with experienced support engineers who can handle questions, assist with service requests, and help troubleshoot systems.

Hours of operation	Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)
Telephone (North America)	877-946-6320, prompt #3, #1
Telephone (International)	1-510-777-7133, prompt #3, #1
E-mail	support@dzsi.com
Web - available 24 x 7 to submit and track field issues/ problem reports	https://dzsi.com/support/#TAC Click DZS Problem Reporting System

If you purchased the product from an authorized dealer, distributor, Value Added Reseller (VAR), or third party, contact that supplier for technical assistance and warranty support.

Hardware repair

If the product malfunctions, all repairs must be authorized by DZS with a Return Merchandise Authorization (RMA) and performed by the manufacturer or a DZS-authorized agent. It is the responsibility of users

requiring service to report the need for repair to DZS Quality & Service as follows:

- Complete the RMA Request form (<https://dzsi.com/support/#RMA>) or contact DZS Quality & Service via phone or email:

Hours of operation: Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)

E-mail: support@dzsi.com (preferred)

Phone: 877-946-6320 or 510-777-7133, prompt #3, #2

- Provide the part numbers and serial numbers of the products to be repaired.
- All product lines ship with a minimum one year standard warranty (may vary by contract). DZS warrants all repairs for 90 days or the remainder of the standard warranty (whichever is greater).
- DZS will verify the warranty and provide the customer with a repair quote for anything that is not under warranty. DZS requires a purchase order or credit card for out of warranty fees.

1

OVERVIEW

This chapter introduces DZMS. It includes the following sections:

- [DZMS system components, page 17](#)
- [About the DZMS database, page 20](#)
- [DZMS system terminology, page 21](#)
- [Sequence of system setup tasks, page 21](#)

DZMS system components

DZS Management System (DZMS) is the network management system that provides service-to-subscriber management for DZS local loop network elements.

DZMS includes a comprehensive suite of applications that deliver fault, configuration, performance, and security management functions as well as operator account management. The DZMS system allows carriers and service providers to operate networks efficiently and to rapidly deploy new services.

DZMS supports a connection with Operations System Support (OSS) gateway to provide a published interface for bidirectional transfer of all management information to the DZS network, thus integrating DZMS applications into service provider back-office systems and third-party applications. OSS gateway modules (with either SOAP/XML or Netconf/Yang) have separate licenses from DZMS.

DZMS is based on a distributed client/server network management architecture. The components in the DZMS environment are:

- The DZMS application server is a UNIX process that provides a suite of management applications to support all functional areas of network management. The server responds to client requests to manage DZS network elements and updates the DZMS database.
- DZMS managers/services are entities that deliver network management services:
 - Configuration services manager (CS) makes configuration changes to network elements and provisions subscribers and end-to-end services within the DZS environment. Supports the use of templates, which allow the operator to streamline configuration tasks. Handles element software downloads.

- Fault services manager (FS) handles traps and alarms from network elements. Forwards traps to destination addresses as needed. Applies rules to determine what action to take on alarms. Trap and alarm information identifies network issues that need to be addressed.
- Performance services manager (PS) tracks network performance data in real time, collects interval statistics, and monitors the status of network elements. This data allows service providers and subscribers to track trends and service levels.
- Administration services manager (ADS) enforces security through operator authentication and authorization. Controls system access by DZMS client operators based on operator login and assigned permissions. Tracks and stores operator actions in an audit log.
- Database service (DS) manages and stores network management data in the DZMS database.
- Configuration Synchronization service (CSS) ensures that network elements, the DZMS database, and DZMS clients remain synchronized.
- DZMS works with either Oracle or MariaDB databases. The database stores fault, configuration, and performance data, as well as subscriber and operator information. Latest versions of DZMS support downloads of MariaDB, Oracle DB will be retained when performing a DZMS upgrade from 3.2.2 or would need to be installed separately.
- NetHorizon client is the Java application that operators use to navigate, monitor, and manage objects in their DZS Management System (DZMS) networks. Objects are networking components such as network devices, cards, physical ports, logical interfaces, and permanent virtual circuits (PVCs).

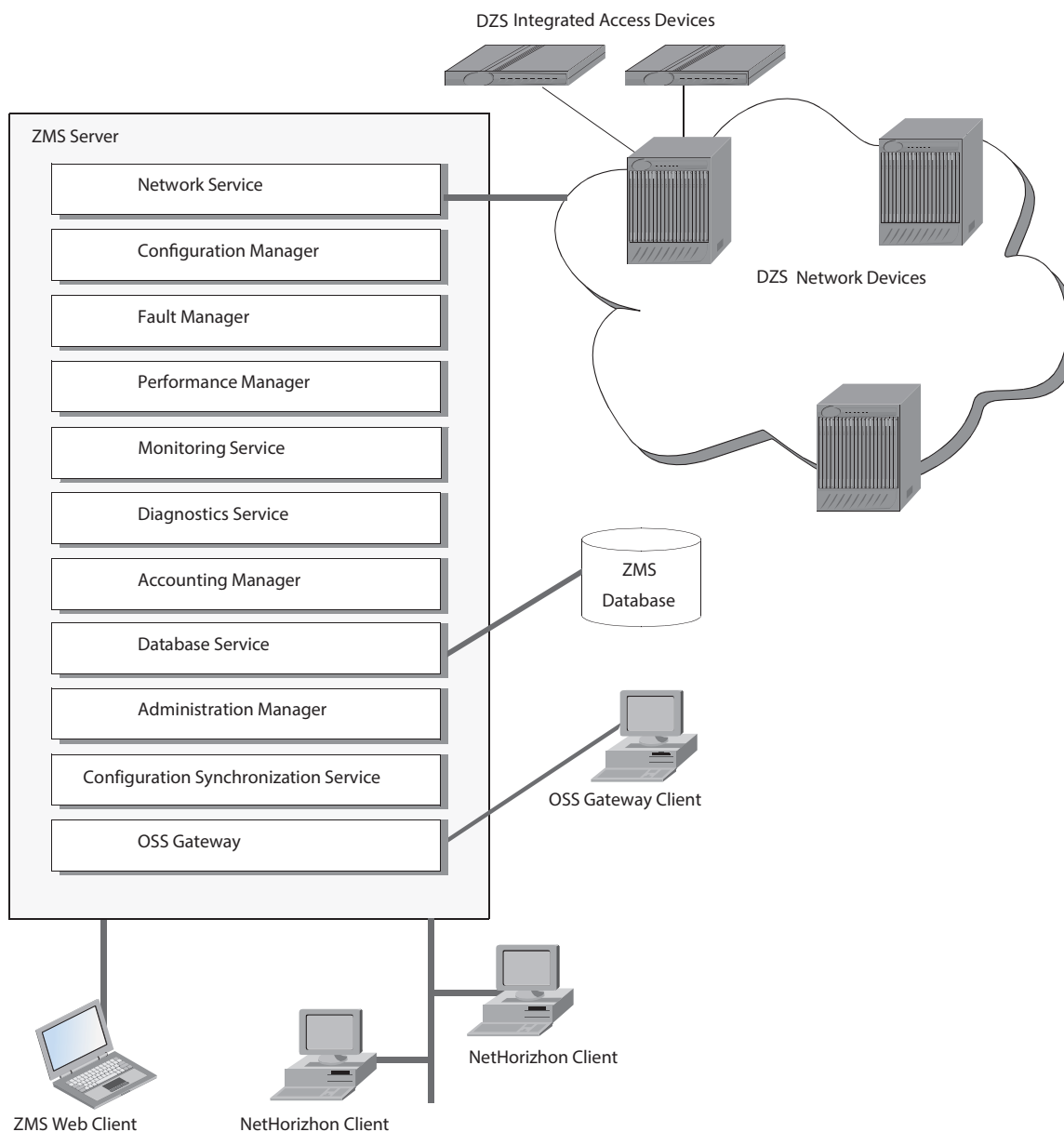
The DZMS administrator determines the specific operations that are offered to operators. If the DZMS administrator provides full DZMS permissions, operators can perform the following types of operations from the NetHorizon application:

- View and monitor the status of network elements.
- Make configuration changes to network elements and provision subscribers and end-to-end services within the DZS SLMS environment.
- View traps and alarms from network elements and configure alarm forwarding to destination addresses, as needed.
- Track real-time and interval network performance statistics.
- Run diagnostics on network elements.
- DZS network objects are DZS network devices and their various components.

- DZMS Web Client is a web browser interface used to manage GPON, ActiveEthernet ports, and VDSL with the latest version. For the most part, the functionality that is available in NetHorizon is available in Web Client. Some functionality has been added to Web Client in the area of fault history and provisioning wizards.

The actual functionality provided to the operator depends on the operator permissions assigned by the DZMS administrator. If an operator does not have permissions to a particular function, menu options that support that function are hidden or restricted within the DZMS clients. For example, if the DZMS administrator does not grant permission for the operator to run diagnostics, NetHorizon hides that function, whereas DZMS Web Client disables access to the function.

[Figure 1](#) illustrates the DZMS system environment.

Figure 1: DZMS environment

About the DZMS database

Either Oracle or MariaDB relational database software is used to store DZMS database information. The software provides access to the full set of database tools, including SQL queries, backup, and recovery of database files.

DZMS maintains a database. For information on administrative tasks you may need to perform on the DZMS database, see [Chapter 4, DZMS database server administration](#)

DZMS system terminology

[Table 1](#) describes the terminology used in the DZMS system.

Table 1: DZMS system terminology

Term	Definition
Admin	The admin user can manage groups and users and grant permissions to the users. The admin user cannot view or manage DZMS objects.
Default group	The default group contains users and objects that were added before any groups were explicitly created.
Group	A group is a collection of users and objects for the purpose of access control.
Manager	A manager is the entity that delivers a network management service by way of the client. For example, the configuration manager delivers the provisioning service so that operators can configure network elements and subscribers. The administration manager enforces security through operator authentication and authorization.
Object	An object is a DZMS network component, such as a network device, card, physical port, logical interface, PVC, and so on.
Operator or user	An operator or user can view and/or manage DZMS objects in his or her group. The permissions associated with an operator's account determines the operations he or she can perform. An operator <i>cannot</i> add groups or users.
Permissions	Permissions are the set of allowed actions that an operator can perform, ranging from add, modify, and delete privileges to view-only privileges for DZMS objects.
Subscriber	A subscriber is the remote user to which the DZS system provides services.
User Administrator	A user administrator can manage users in his or her group and grants permissions to the users. The user administrator can also manage DZMS objects in his or her group, but <i>cannot</i> add groups.

Sequence of system setup tasks

[Table 2](#) shows the proper sequence for system setup tasks.

Table 2: Sequence for setup operations

Setup operation	See
Install DZMS system components	<i>DZMS Installation Guide</i>
Administer DZMS system components	DZMS system components, page 17.
Create groups and operator accounts and assign permissions	Administering Operator Accounts, page 59.

2

SYSTEM ADMINISTRATION

This chapter describes how to administer the various components of the DZMS system. It describes the following administrative tasks:

- [Starting, Stopping, and Checking the status of DZMS Server, page 23](#)
- [Configuring DZMS managers \(properties\), page 23](#)



Note: For instructions on installing the DZMS system, see the *DZMS Release Notes* that shipped with this release.

Starting, Stopping, and Checking the status of DZMS Server

To start, stop or check the DZMS status, use the following commands:

```
<sudo> systemctl [start|stop|status] dzms
```

Note that command text in square brackets means choose one of the options. e.g. `[start|stop|status]`. Command text in angle brackets means text is optional. e.g. `sudo` is optional if logged in as oracle user, it is needed if not logged in as root user.

Configuring DZMS managers (properties)

The following services with properties are supported:

- [Updating Properties](#)
- [Network service](#)
- [Fault manager](#)
- [Monitoring service](#)
- [Configuration synchronization service](#)
- [Configuration service](#)
- [Asynchronous service](#)
- [Debug service](#)
- [MTAC external relay alarm templates](#)

- [ONU/ONT serial number format \(Hexa-decimal, Decimal\)](#)
- [Display ONT/ONU MAC address](#)
- [Enable wildcard searches](#)
- [Hide the Find Object menu item](#)
- [RADIUS client support in DZMS](#)
- [LDAP client support in DZMS](#)
- [AD client support in DZMS](#)
- [Trap Multi-Threading](#)
- [OSS Returns “ProfileType_NOT_FOUND_ERROR” If No Records Found](#)
- [Allow Downlink Bridges](#)
- [Configure Alpha LPS-36](#)
- [Configure Eltek Smartpack Alarm Reporting](#)

Updating Properties

- [Updating DZMS Properties in the Database, page 24](#)
- [Updating the remaining DZMS Properties, page 25](#)

Updating DZMS Properties in the Database

The properties within the following property files, which used to be stored in */opt/zms/lib/com/zhone/zms/properties* directory, are now being stored in the ZMS Database:

ADS.properties

CS.properties

CSS.properties

NWS.properties

MS.properties

FaultService.properties

DumpRestore.properties

Async.properties

NetHorizon.properties

Radius.properties

WebClient.properties

The properties in the above property files must now be updated from either NH client or DZMS web client as admin user.

All property changes take effect immediately, except for properties within these two categories:

- For changes to Fault Service properties, users will need to restart DZMS application.
- For changes to NetHorizon properties, users will need to restart their NH Client sessions.

To update the properties in DB from NH client or DZMS Web Client, use the following procedures:

1 In NH client:

- a** Log into the NH client as admin user.
- b** In the NH Client, select **ZMS Properties** in the **Tools** pull-down menu.

The **Modify ZMS Properties** window appears.

- c** From the **Modify ZMS Properties** window, you can select ZMS Property Category, and change the property values of property attributes.

2 Or in DZMS Web Client:

- a** Log in the DZMS Web Client as admin user.
- b** Click on **Administration** then ZMS Properties tab.

The **ZMS Properties** window appears.

- c** From the **ZMS Properties** window, you can select ZMS Property Category, and change the property values of property attributes.

Updating the remaining DZMS Properties

The following property files located in */opt/zms/lib/com/zhone/zms/properties* are not moved to the Database:

UserDefined_template.properties

UDAlarmSeverity_template.properties

UDAlarmDescription_template.properties

FSAlarmSeverity.properties

FSAlarmDescription.properties

Debug.properties

Debug.client_template.properties

To change the remaining properties, edit the file and restart DZMS application, except *Debug.properties* can be done via the following URL.

DZMS version 3.3 and prior versions:

<http://zms-hostname:7001/zms/Debug>

ZMS version 3.4 and later versions:

`http://zms-hostname:8080/zms/Debug`

DZMS version 4.x and later versions:

`http://dzms-hostname:8080/dzms/Debug`

Network service

To configure the network service, modified network service variables from NetHorizon / Web Client. Network service variables are listed in [Table 3](#).

Table 3: Network service (NWS properties) variables

Variable	Default value	Description
DefaultCpeCommunityString	ZhonePrivate	The CPE default community string.
MaxVarBindsToPoll	16	Devices at times fail to respond if too many OIDs are requested at a time in the same snmpget command. Hence defaulted to 16. If there are 40 OIDs to be polled, DZMS server will divide them into 3 requests with 16+16+8 OIDs in each, and consolidate the response before it sends to client.

Table 3: Network service (NWS properties) variables (Continued)

Variable	Default value	Description
mibsToLoad	SLMSMibs SkyZhoneMibs	Determines which mib groups to load. Removing unused mib groups will decrease startup time. Those mibs are located on the DZMS server in directory <i>/install directory/zms/lib/mibs</i>, e.g. <i>/opt/zms/lib/mibs</i>. Possible mib groups which are defined in <i>mibs.properties</i> are: commonMibs - RFC and standardized mibs This is always needed and has to be first in list. SLMSMibs - mibs specifying SLMS mibs (MxK, Malc, Raptor, ...) GigamuxMibs- mibs specifying GigaMux mibs IPDMibs - mibs for IPD 4000/12000/8400 ... GM6400Mibs - GigaMux 6400 device mibs VPSMibs - Valence Power System mib BSMibs - HP BitStorm device mibs Verso8300Mibs - Verso 8300 device MIBs SkyZhoneMibs - SkyZhone device MIBs ALPSMibs - Alpha LPS-36 device MIBs (To configure Alpha LPS-36, use the section Configure Alpha LPS-36 on page 57) EltekSP2Mibs - Eltek Smartpack device MIBs (To configure Eltek smartpack alarm reporting, use the section Configure Eltek Smartpack Alarm Reporting on page 57) Example value for when only SLMS products exist in network: mibsToLoad=commonMibs SLMSMibs
PollWakeUp	1000	LD and statistics poller interval (milliseconds)
SNMPCommString	ZhonePrivate	The network element default community string
SNMPRemotePort	161	SNMP default remote port.
SNMPRetries	1	Request retries: number of SNMP retries the server attempts before it declares a timeout. Note: With each retry, the server increases the request timeout value by a factor of three.
SNMPRetriesForCPE	1	Sets number of retries before timeout. The timeout value grows exponentially for each retries. Timeout value is doubled for each retry. For example, if the timeout is set to 5000 (meaning 5 seconds) and retries is set to 2, the first retransmission will happen after 5 seconds, the second after 15 seconds etc.
SNMPStartLocalPort	161	SNMP default local.

Table 3: Network service (NWS properties) variables (Continued)

Variable	Default value	Description
SNMPTimeout	8000	Request timeout: base amount of time, in millisecond increments, between SNMP retries from the server to the DZS device. With each retry, the server increases the timeout value by a factor of three.
SNMPTimeoutForCPE	20000	Number of milliseconds for CPE to wait before SNMP command times out.
SNMPWriteCommString	DZSPRivate ZhongPrivate	SNMP Read-Write community name.

Fault manager

The following properties files provide the flexibility to customize DZMS alarm handling and processing.

- `FSAlarmSeverity.properties` file
- `FSAlarmDescription.properties` file
- Fault manager variables in the NetHorizon / Web Client

To configure the fault manager, set the variables in the following files / GUI:

- Fault manager variables in NetHorizon or Web Client. Fault manager variables are listed in [Table 4](#). You can customize the settings from NetHorizon/ Web Client GUI for email notifications, trap/alarm forwarding, trap storm, alarm history log and other fault service settings.
- *FSAlarmSeverity.properties*. Alarm severity levels can be changed by modifying the *FSAlarmSeverity.properties* file. Alarm severities that are commented out in the *FSAlarmSeverity.properties* (with a “#” sign) derive their severity level from a trap. Alarm severities that are not commented in the *FSAlarmSeverity.properties* file have the severity level defined in the file. This severity assignment overrides the trap severity setting on the MALC to one of the severity settings (Critical, Major, Minor, Warning or Informational). Refer to [DZMS Legacy Traps and Alarms on page 119](#) for the SNMP traps that generate alarms.

For more information about how to modify alarm severities, see [Appendix C, Modifying DZMS alarm severities and descriptions, on page 125](#)

- *FSAlarmDescription.properties*. You can customize individual alarm descriptions to include one or more attribute values. To add an attribute to an alarm description, use the format `$$attribute_name$$`, where *attribute_name* is an existing attribute on the object generating the alarm. This description is appended to the contact name assigned to the alarm relay contact through the CLI or DZMS. For example, to identify what particular object generated an alarm, you can add the object name to the alarm description:

The status of AAL type 2 PVC (\$\$name\$\$) has gone down

For more information about how to modify alarm descriptions, see [Appendix C, Modifying DZMS alarm severities and descriptions](#), on page 125.

Table 4: Fault manager (FaultService properties) variables

Variable	Default value	Description
alarmForwardLogFile	<i>AlarmForward.log</i>	Name of the log file to capture forwarded alarms.
alarmForwardPort	162	Server port on which alarms are received.
alarmForwardStopList	UpdateFailed UpdateCompleted PartialUpdateCompleted	Alarm(s) to be excluded from forwarding to specified host(s) and excluded from notification mail messages.
alarmForwardWhiteList	—	Used when alarms are being forwarded. If value is not blank, only alarms specified in this list will be forwarded. Multiple alarm names can be specified in a comma separated list.
alarmGmailOAuthClientId	—	oauthClientId got from google developer console.
alarmGmailOAuthRefreshToken	—	RefreshToken got from google developer console.
alarmGmailOAuthSecret	—	oauthSecret got from google developer console
alarmLogFile	<i>alarm.log</i>	Name of the log file to capture alarms.
alarmMailEnableTLS	false	If true smtp will use TLS.
alarmMailPassword	—	Mail sender password.
alarmMailRetryCount	10	Number of times mail will be retried, if there is a problem.
alarmMailRetryInterval	30000	Interval (in milli-seconds) at which mail will be retried.
alarmMailSender	—	Name or e-mail address of the sender of the alarm notification mail messages (for example “ZMS Fault Manager” or jsmith.xyz.com.). If the name contains a blank character, enclose the name in quotation marks.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
alarmMailServer	—	Name of the mail server that manages mail for the individuals to receive mail message notification of alarms (for example email.ny.xyz.com.)
alarmMailServerPort	25	Server port on which the mail server that manages mail for the individuals to receive mail message notification of alarms.
authPassword	—	Authentication password.
authProtocol	MD5	Authentication protocol.
createFiberCutAlarms	false	When enabled, this property allows a user to enter ONU splitter information. If all ONUs on the splitter are down, a fiber cut alarm is created.
createOntConnectedAlarm	false	If this value is true, DZMS will create an alarm each time a new GPON ONT is connected.
criticalAlarmForwardHosts	—	Alarm forward host(s) to receive critical alarm information. Use the format: <IP_address> <community_name> If you omit the community name, DZMS uses the default community name ZmsTrapPrivate. Separate the entries by commas.
criticalAlarmMailRecepients	—	E-mail address of the recipients of the critical alarm notification mail messages (for example jsmith.xyz.com.) Separate the addresses by commas.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
dontDropIfNameContains	—	Defines whether or not a trap gets dropped before the standard processing begins. If the name of a trap is specified in trapsToDrop, all traps with that name will be dropped unless the trap has an ifAlias varBind and ifAlias contains one of the values in "DontDropIfNameContains". This feature was mainly added to drop zhongGponOnuError traps on ONUs that do not have VIP or GSM in their name. It can be a list of comma separated items. ex. dontDropIfNameContains=VIP , GSM
enableAutoAlarmStateUpdates	true	Setting enableAutoAlarmStateUpdates to false prevents DZMS from sending alarm state updates every time an alarm is added/deleted/modified. The Update From Database menu option in the NetHorizon still can retrieve alarm state.
enableClearAlarmCheck	true	Enables or disables the ability to clear alarms if the condition still exists on the device. If set to true, DZMS will check whether the alarm exists before clearing it. If set to false, DZMS will not check whether the alarm exists before clearing it.
enableTrapSequencer	false	Set to true to enable trap sequencer. Set to false to disable trap sequencer.
engineID	—	SNMP v3 engine ID.
enhancedAlarmHistoryEnabled	false	Setting to true stores alarm history in the DB. In order for this to work, the Warehouse package needs to be installed.
fetchZmsDiskSpaceInterval	10	The interval in minutes for fetching DZMS disk space. Setting this property to "0" will disable fetching disk space.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
forwardAllAlarmsDay	none	DZMS can be configured to re-forward all existing alarms. DZMS will not forward all alarms if value equals "none". Use "all" to forward daily
forwardAllAlarmsTime	04:00	The time to re-forward all existing alarms. Format = HH:MM. Used with forwardAllAlarmsDay.
informationalAlarmForwardHosts	—	Alarm forward host(s) to receive alarm information. Use the format: <IP_address> <community_name> If you omit the community name, DZMS uses the default community name ZmsTrapPrivate. Separate the entries by commas.
informationalAlarmMailRecepients	—	E-mail address of the recipients of the informational alarm notification mail messages (for example jsmith.xyz.com.) Separate the addresses by commas.
keepAliveTrapInterval	600	Interval for posting a trap that DZMS server is Up (in seconds).
mailClearedAlarms	false	Enables or disables the ability to mail cleared alarms to all members listed in zmsAlarmMailRecepients. If sets to true, and if the properties alarmMailServer, alarmMailSender, and zmsAlarmMailRecepients are set to non-empty values, and the alarm that is cleared is not part of alarmForwardStopList, the cleared alarm will mailed to the all members listed in zmsAlarmMailRecepients.
majorAlarmForwardHosts	—	Alarm forward host(s) to receive major alarm information. Use the format: <IP_address> <community_name> If you omit the community name, DZMS uses the default community name ZmsTrapPrivate. Separate the entries by commas.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
majorAlarmMailRecepients	—	E-mail address of the recipients of the major alarm notification mail messages (for example jsmith.xyz.com.) Separate the addresses by commas.
maxAlarmLogRecords	5000	Specifies the maximum number of alarm log records stored in the alarm history log. When the maximum is reached, additional log records are not recorded.
minorAlarmForwardHosts	—	Alarm forward host(s) to receive minor alarm information. Use the format: <IP_address> <community_name> If you omit the community name, DZMS uses the default community name ZmsTrapPrivate. Separate the entries by commas.
minorAlarmMailRecepients	—	E-mail address of the recipients of the minor alarm notification mail messages (for example jsmith.xyz.com.) Separate the addresses by commas.
NetGuardianCriticalPowerAlarm1_Name	AC Fail	DZMS external relay alarms that has the one of the two names will be treated as NG critical power alarm. The values can be changed if needed to be and restart the DZMS server to make the changes effective.
NetGuardianCriticalPowerAlarm2_Name	Multiple Rectifier Fail	DZMS external relay alarms that has the one of the two names will be treated as NG critical power alarm. The values can be changed if needed to be and restart the DZMS server to make the changes effective.
NetGuardian_IP_1 NetGuardian_IP_2 NetGuardian_IP_3 NetGuardian_IP_4 NetGuardian_IP_5	—	DPS/MarkTwain alarm bit settings on NetGuardian device. At least one of the NetGuardian_IP must be set to a valid NG IP for being able to set the alarm bits on NG devices.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
NetGuardian_Region_1 NetGuardian_Region_2 NetGuardian_Region_3 NetGuardian_Region_4 NetGuardian_Region_5	—	This Setting is optional and maps the DZMS regions to NetGuardian devices. By default, the NG bits will be set at DZMS level. If devices of a particular region need to go to a specific NG, all regions must be set in these attributes. If NG1 takes 3 regions, names of those 3 regions can be set to NetGuardian_Region_1 as comma separated list of values. If no NG is specified for a region, first available NG will be used for setting the alarm bit.
NetGuardian_SnmpSet_Community_1 NetGuardian_SnmpSet_Community_2 NetGuardian_SnmpSet_Community_3 NetGuardian_SnmpSet_Community_4 NetGuardian_SnmpSet_Community_5	dps_public - - - -	The default community name for snmp sets. But this can be changed to a different value if needed. At least one of the following must be non-empty string for being able to set the alarm bits on NG devices.
numberOfTrapProcessingThreads	1	Specifies the number of trap threads can be used to process traps. The largest recommended value is 20% of the server's cores. For example: An appropriate value for a server with 32 cores would be between 1 and 6.
NumDaysToKeepAlarmHistory	30	If EnhancedAlarmHistory is being used, NumDaysToKeepAlarmHistory specifies how many days of alarm history to store.
numTrapsToEnableThrottling	800	Specifies how many traps from the same device need to be in queue before initiating trap throttling.
privPassword	—	The properties: trapReceiverPort, username, engineID, authProtocol, authPassword, privPassword, are used by trap receiver. These properties determine which traps this fault service can receive.
showOnuAlarmsForNetwork	true	If value equals true, ONU and ONT alarms will not be shown in the alarm viewer when alarms are viewed for the entire network.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
trapCacheSize	250	Specifies the number of traps cached by DZMS. If cache is full and DZMS receives more traps, those traps will be dropped by DZMS. Because increasing the cache size will have memory and performance implications, use caution when modifying this value.
trapForwardLogFile	<i>TrapForward.log</i>	Name of the log file to capture forwarded traps.
trapForwardPort	162	Port of the destination host to which traps are forwarded.
trapForwardStopList	zhoneTrapConfigSyncChange	Trap(s) to be excluded from forwarding to specified host(s).
trapForwardWhiteList	—	Used when traps are being forwarded. If value is not blank, only traps specified in this list will be forwarded. Multiple trap names can be specified in a comma separated list.
trapLogFile	<i>trap.log</i>	Name of the log file to capture traps.
trapProcessingQueueMaxSize	1200	The properties: useTrapThrottling, trapProcessingQueueMaxSize, numTrapsToEnableThrottling, trapThrottlingPeriod are used to tweak the trap throttling feature. All traps enter the trapProcessingQueue while waiting to be processed. Once the queue is full, additional traps will be dropped.
trapReceiverPort	162	Server port on which traps are received.
trapSeqTimeout	30000	Timeout (in milliseconds) before the fault manager requests the DZS device resend a missing trap.
trapSeqTrapWindowSize	5	Number of higher-numbered traps received after a missing trap before the fault manager requests the DZS device resend the trap.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
trapsToDrop	—	Defines whether or not a trap gets dropped before the standard processing begins. If the name of a trap is specified in trapsToDrop, all traps with that name will be dropped unless the trap has an ifAlias varBind and ifAlias contains one of the values in "DontDropIfNameContains". This feature was mainly added to drop zhongGponOnuError traps on ONUs that do not have VIP or GSM in their name. It can be a list of comma separated items. ex. trapsToDrop=zhongGponOnuError,zhongGponOmciOnuAlarmsTrap
trapStormThreshold	50	trapStormThreshold, and trapStormTimeout are used for handling trap storms. trapStormThreshold is number of traps queued by DZMS to identify that a device is in the middle of a trap storm.
trapStormTimeout	60	Specifies the period of time, in minutes, that DZMS waits before declaring an end to a trap storm. See the description for trapStormThreshold. While in trap storm mode and waiting for timeout, you can manually end the timeout period by right-clicking a device in the NetHorizon and selecting Update Display from Database .
trapThrottlingPeriod	15	Specifies the length, in minutes, that DZMS will throttle traps from offending device.
useOltAlarmCorrelation	false	This hides OLT linkDown if no ONUs are provisioned and changes OLT linkDown to minor if last ONU went down because of a dying gasp.
username	—	SNMP v3 username.
useTrapThrottling	true	Setting "useTrapThrottling" to "true" will enable Trap Throttling and automatically override Trap Storm settings.

Table 4: Fault manager (FaultService properties) variables (Continued)

Variable	Default value	Description
warningAlarmForwardHosts	—	Alarm forward host(s) to receive warning alarm information. Use the format: <IP_address> <community_name> If you omit the community name, DZMS uses the default community name ZmsTrapPrivate. Separate the entries by commas.
warningAlarmMailRecepients	—	E-mail address of the recipients of the warning alarm notification mail messages (for example jsmith.xyz.com.) Separate the addresses by commas.
zmsAlarmMailRecepients	—	If DZMS login failure alarm raising is enabled. Every time user fails on the login, a DZMS login failure alarm will be raised, and an alarm notification email with username and host IP information will be sent out if alarmMailServer, alarmMailSender, and zmsAlarmMailRecepients are specified in the property file.
zmsDiskSpaceCriticalThreshold	10	The critical "usableSpace/totalSpace" percentage ratio threshold
zmsDiskSpaceMajorThreshold	20	The major "usableSpace/totalSpace" percentage ratio threshold
zmsDiskSpaceMinorThreshold	30	The minor "usableSpace/totalSpace" percentage ratio threshold

Monitoring service

To configure the monitoring service, set the variables using DZMS Web Client or NetHorizon Client. Monitoring service variables are listed in [Table 5](#).

Table 5: Monitoring service variables

Variable	Default value	Description
PollInterval	120	Frequency of poll requests, in seconds, made to the network to determine if the device is reachable.

Table 5: Monitoring service variables (Continued)

Variable	Default value	Description
PollRetries	0	The number of times ZMS will attempt to reach the device before declaring the device unreachable.
PollTimeout	5	The number of seconds before an SNMP poll request to a device timeouts.

Configuration synchronization service

To configure the configuration synchronization service, set the variables using DZMS Web Client or NetHorizon Client. Configuration synchronization service variables are listed in [Table 6](#).

Table 6: Configuration synchronization service (CSS property) variables

Variable	Default value	Description
addCpeTrapDestinations	true	In lab environments where there are multiple DZMS servers, addCpeTrapDestinations needs to be changed to false. If not, adding a MXK to a new DZMS will change the trap address on all the
continueOnFailure	true	This property can be set to notify CSS to continue processing updates even if database modifications fail. The failures will be reported to the user as alarms.
continueSchedCfgSyncEvenIfThresholdIsExceeded	true	CpuThresholdPercentage and continueSchedCfgSyncEvenIfThresholdIsExceeded are used to generate a warning dialog whenever the current CPU usage is $\geq$ the following setting to indicate that device is busy and config sync is better to be scheduled later. This takes a max value of 100. Even if the value is greater than 100, it will be treated as 100.
CpuThresholdPercentage	75	CpuThresholdPercentage and continueSchedCfgSyncEvenIfThresholdIsExceeded are used to generate a warning dialog whenever the current CPU usage is $\geq$ the following setting to indicate that device is busy and config sync is better to be scheduled later. This takes a max value of 100. Even if the value is greater than 100, it will be treated as 100.
deleteProcessedUpdateFiles	true	Set to true if update files should be deleted upon completion. NOTE: Update files that are not successfully processed will not be deleted.

Table 6: Configuration synchronization service (CSS property) variables (Continued)

Variable	Default value	Description
disablePartialCss	false	If true, DZMS will ignore partial config sync information. This should only be used as a temporary work-around if partial config syncs are causing issues.
ftpUserName	zmsftp	The username CSS will inform the device to use in order to ftp the update file to DZMS.
initialPollWait	1000	The number of milliseconds CSS will wait to poll syncStatus for the first time after an update request.
locateFullUpdateRetry	10000	Retry time for locating the update file on the DZMS machine. CSS will wait this amount of milliseconds between polling for the configSyncStatus before retrying. This timeout is for full updates.
locateUpdateRetry	1000	Retry time for locating the update file on the DZMS machine. CSS will wait this amount of milliseconds between polling for the configSyncStatus before retrying. This timeout is for partial updates..
numberOfRecordsBetweenPauses	100	This parameter is used in conjunction with pauseToAllowOtherThreadsToRun. It specifies how many records to process in between pauses.
numFullUpdateStatusPolls	200	The number of times the device will be polled for the update file transfer status. This is the number of polls that will be attempted before CSS will timeout for full updates. The current settings of 24 polls spaced 5 seconds apart (see locateUpdateRetry) allow for a two-minute timeout..
numUpdateStatusPolls	100	The number of times the device will be polled for the update file transfer status. This is the number of polls that will be attempted before CSS will timeout for partial updates. The current settings of 24 polls spaced 5 seconds apart (see locateUpdateRetry) allow for a two-minute timeout.
offline	false	If testing offline, "offline" should be set to true and "offlineUpdateFile" should be set to the name of the update file to be processed.

Table 6: Configuration synchronization service (CSS property) variables (Continued)

Variable	Default value	Description
partialConfigSyncWait	10000	Number of milliseconds DZMS will wait to send a config sync request to device after receiving a partial config sync trap. Device has a lock mechanism so that only one host (CLI, DZMS, DZMSLT or third-party SNMP) can do sets to device at any given time. The lock has a timeout period of 10 seconds. DOMCI actions cause partial config syncs. These actions do not have the 10 second lockout. This value has been decreased so that it does not take as long to add all of the DOMCI records.
pauseToAllowOtherThreadsToRun	0	While config sync is running, other requests are not executed. To resolve this issue, pauses have been added into the config sync code. This parameter specifies the number of milliseconds to pause each time a pause is executed.
updateCompletedDesc	A full Config Sync update has completed successfully	The descriptions used for the alarms generated upon completions and failures of CSS updates.
updateFileLocation	/opt/zmsftp/	The location from which CSS will retrieve the update file. NOTE: Directory location MUST have trailing "/" For example, "/opt/zmsftp/" instead of "/opt/zmsftp".
useDevicePortDescriptions	true	If this value is true, during the config sync of the device from DZMS the physicals description values will be updated on the DZMS database. The old values (1.15 and less than release physicals values will be lost).

Configuration service

To configure the configuration service, set the variables using DZMS Web Client or NetHorizon Client. Configuration service variables are listed in [Table 7](#).

Table 7: Configuration service (CS property) variables

Variable	Default value	Description
auto-delete-cc-pvcs	true	When an ATM Cross Connect is deleted, this determines whether the pvcs # that it cross connects should be deleted automatically.
auto-delete-related-voice	true	This flag determines whether items related to voice connections are deleted when the voice connection is deleted, i.e.: Cids, Crvs, Vcls.
cpeMgrGtpIndex	1100000001	The GTP index that will be used by DZMS GPON ports for ONT "Launch Web UI" command.
cpeMgrNgponGtpIndex	1200000001	The GTP index that will be used by DZMS NGPON ports for ONT "Launch Web UI" command.
csPollInterval	5000	Poll interval for finding out whether the poll (example : software download or restore) operation has been completed in msec. 5000 msec is 5 secs.
csPollNumOfTimes	200	Max number of times to poll. The total time for polling is equal to $csPollInterval * csPollNumOfTimes$.
dynamicCpeMgrTimeOut ValueInMinutes	1440	Controls how long the cpeMgr connection stays up after the "launch CPE Web UI" command is issued.
maxBridgesAllowed	0	Maximum number of bridges that can be created per device. A check for max bridges should not be necessary. A value of 0 causes DZMS to skip the max-bridges check. The last value was 12288. To re-enable this check to set the value back to 12288..
maxNumOfFindAddressThreads	10	Maximum number of devices that DZMS will query for a MAC/IP address at the same time. Increasing this value will speed up address searches. It will also increase the computing resources required to execute the search.
modifyFullSerialNo	false	The MXK is not allowing serialNoVendorId to be set to an empty string. Setting serialNoFull to an empty string, sets serialNoVendorId and serialNoVendorSpecific to empty strings. If this flag is set to true, serialNoVendorId will not be modified when serialNoFull is set to an empty string.
mtacTestTimeout	300000	MTAC test timeout. By default, DZMS will wait for the old test to complete for 5 minutes and then cancel the old test and continue with the current test. Time in milliseconds.

Table 7: Configuration service (CS property) variables (Continued)

Variable	Default value	Description
omciDnldPollNumOfTimes	40	Max number of times to poll for an OMCI profile download. The polling interval is .5 seconds. Max number of times to poll for an OMCI profile download. The polling interval is .5 seconds. The total time for polling is equal to $\text{omciDnldPollNumOfTimes} * 0.5$ seconds.
performSerialNumberUnique Check	true	To disable network wide serial number uniqueness check, change value to false.
setCCnameOnDevice	false	Starting in 1.12.2, the device can store the cross-connect name as handleId. The handleId field is only 16 chars, and DZMS has always allowed 32 char CC name. The handleId field was added just for a specific customer and they agreed to a 16 char handleId. However, we don't want to force other customers to shorten all of their cross-connect names.
showExtraBridgeParameters	false	Sends FilterBroadcast, CustomArp and FloodMulticast values to the MXK instead of allowing the MXK to determine the values based on bridgeType. If this value is modified, the same value in NetHorizon.properties needs to be modified also.
showOssNoProfErrorOnZero Records	false	Setting this to true will show ProfileType_NOT_FOUND_ERROR whenever there are 0 records returned as part of get / list response. Example => Running list on Device will return Device_NOT_FOUND_ERROR if list on Device Filter_Type did not return any records. Also, the following flag will only work when Handle_Type and Hadnle_Id combination is skipped. Example => Filter_Condition=Device_Id=100 shall be instead of specifying Handle_Type=Device and Handle_Id=100.
sqlQueryTimeoutInSeconds	60	If an SQL query takes longer than the value here, a timeoutException will be thrown. A value of "0" disables the timeout timer.
SshProxyFeatureEnabled	false	Proxy process feature enable, SSH proxy process to device once value set as true else no proxy.
SshProxyStartPort	10000	ssh proxy default port is "10000"
SshProxyTimeOutValueIn Minutes	60	Controls how long the ssh proxy connection stays up after the "SSH To Device" command is issued.

Table 7: Configuration service (CS property) variables (Continued)

Variable	Default value	Description
use-alternate-bridge-name	false	Bridges on top of bond groups create the bridge name by appending "-n2nbond-vlan" to the physical (bond group) name. Since the bridge name has to be ≤ 35 chars, the max bond group name is 22 chars. In order to allow longer bond group names, an alternate bridge name format can be used. The alternate bridge name is physicalName-xx. Where xx is 01-99. DZMS will ensure that the bridge name is unique.
useOldSspFormat	false	This attribute is added for a customer whose OSS code would break if the SSP (shelf-slot-port) format is changed.
useOnuFsanSN	true	There are two ONT serial numbers in the gpon-olt-onu-config profile. By default, DZMS uses serial-no-vendor-specific-fsan (hex). To instruct DZMS to use serial-no-vendor-specific (decimal), change "useOnuFsanSN" to false.
usingOssHideGponPortLg Change	false	Some customers found all the bridges on a GEM port using bridge.linegroup = gponPort.linegroup. Now that gponPort.linegroup = OLT.linegroup, this does not work. Setting this flag allows them not to change their OSS client. This change only affects OssXml.

Asynchronous service

To configure the asynchronous service, set the variables using DZMS Web Client or NetHorizon Client. Async variables are listed in [Table 8](#).

Table 8: Async service (Async property) variables

Variable	Default value	Description
maxConcurrentHeavyWeightTasks	3	Maximum number of heavy weight tasks to run at the same time. Heavy weight tasks are tasks that use significant DZMS (DB or Java) resources. Current heavy weight tasks are config sync.
maxConcurrentLightWeightTasks	10	Maximum number of light weight tasks to run at the same time. Light weight tasks are tasks that do not use significant DZMS (DB or Java) resources. Current light weight tasks are image download and device backup.

Debug service

To configure the debug service, set the variables in the file *Debug.properties*. Debug service variables are listed in [Table 9](#).

Table 9: Debug service (Debug.properties) variables

Variable	Default value	Description
AdminService	1	Debug setting on admin service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info.
AsyncService=2	2	Debug setting on async service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info.
CommonService	1	Debug setting on common service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info.

Table 9: Debug service (Debug.properties) variables (Continued)

Variable	Default value	Description
ConfigService	1	Debug setting on config service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
ConfigSyncService	2	Debug setting on config sync service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
DataService	1	Debug setting on data service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
PerformanceService	1	Debug setting on performance service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
NetworkService	2	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
DiagService	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
MonitoringService	2	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info

Table 9: Debug service (Debug.properties) variables (Continued)

Variable	Default value	Description
FaultService	2	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
EventManager	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
LeaseManager	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
OSS	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
OSSXML	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
DumpRestoreService	1	Debug setting on network service: 0 - Debug Off 1 - Exception, minimum output 2 - Method, in which method 3 - Info, detailed info
addTime	1	Additional Debug flags 0 - Turn Flag off 1 - Turn Flag on
addThreadName	1	Additional Debug flags 0 - Turn Flag off 1 - Turn Flag on

Table 9: Debug service (Debug.properties) variables (Continued)

Variable	Default value	Description
showExceptions	1	Additional Debug flags 0 - Turn Flag off 1 - Turn Flag on
errsToSysErr	0	Additional Debug flags 0 - Turn Flag off 1 - Turn Flag on
flushTime	1000	Time in milliseconds between debug buffer file writes.
errorFile	ZMSErrors.log	Filenames for errors and debugging.
debugFile	debug.log	Filenames for errors and debugging.
logDir	/weblogic	This directory name must contain the full path, and must use standard # URL style forward slashes to represent directories. e.g.: (NT) c:/zhone or (Unix) /weblogic The dos style (c:\zhone) is not standard URL syntax, so it will not work

MTAC external relay alarm templates

The following properties files provide the templates to customize MXK Test Access Cards (MTAC) external alarm handling and processing.

- *UDAlarmSeverity_template.properties*
Refer to [Changing the normal relay state to close on page 48](#) for the detail.
- *UDAlarmDescription_template.properties*
Refer to [Changing the alarm severity of external relay alarm on page 48](#) for the detail.
- *UserDefined_template.properties*
Refer to [Changing the alarm description of external relay alarm on page 49](#) for the detail.

Follow the instructions and examples in these property files to modify the entries and variables. And also note that:

- MTAC external relay alarm properties files are not used until “_template” is removed from their names.
- The “#” symbol in the properties files means comment. Any lines that start with “#” are ignored.

Changing the normal relay state to close

In order to create an alarm for an external input, MALC must know the normal state (open or closed) of the relay. DZMS allows the normal relay states to be defined in the *UserDefined_template.properties*. By default, the normal state is open.

- If the normal state is defined as “open”, an alarm will be generated when the relay transitions to a close state. The alarm will be cleared when the relay transitions back to open.
- If the normal state is defined as “close”, an alarm will be generated when the relay transitions to an open state. The alarm will be cleared when the relay transitions back to closed.

It is possible to define the normal state on a device or on the entire network. For example:

- To change the normal state of relay with Relay ID 2 on the MALC with IP address 100.100.100.100, enters this entry into the properties file:
`EXTERNAL_RELAY_2_NORMAL_STATE.100.100.100.100=closed`
- To change all relays with Relay ID 2 on the entire network, enters:
`EXTERNAL_RELAY_2_NORMAL_STATE=closed`

All entries that specify an IP address take precedence over entries without an IP address. For example, if both the following entries exist:

```
EXTERNAL_RELAY_3_NORMAL_STATE=closed
EXTERNAL_RELAY_3_NORMAL_STATE.100.100.100.100=open
```

Relay 3 on the 100.100.100.100 device will have the open normal state. And relay 3 on all other devices will have the close normal states.



Note: Existing alarms are not automatically re-evaluated. Perform an update display from database from the device node to update the current alarms.

Changing the alarm severity of external relay alarm

MTAC external relay alarm severity levels (Critical=1, Major=2, Minor=3, Warning=4 or Informational=5) can be changed by modifying the *UDAlarmSeverity.properties* file. By default, the MTAC relay alarm severity level is 4. Refer to [DZMS Legacy Traps and Alarms on page 119](#) for the traps that generate MTAC relay alarms.

You can change the MTAC relay alarm severity level on a device or on the entire network. For example:

- To change the alarm severity of all relay with Relay ID 2 on the MALC with IP address 100.100.100.100, enters this entry into the properties file:
`EXTERNAL_RELAY_2_ALARM.100.100.100.100=3`
- To change all relays with Relay ID 2 on the entire network, enters:

```
EXTERNAL_RELAY_2_ALARM=1
```

All entries that specify an IP address take precedence over entries without an IP address. For example, if both the following entries exist:

```
EXTERNAL_RELAY_3_ALARM=3
```

```
EXTERNAL_RELAY_3_NORMAL.100.100.100.100=1
```

A relay 3 alarm on the 100.100.100.100 device will be critical. And a relay 3 alarm on all other devices will be minor.



Note: Existing alarms are not automatically re-evaluated. To change the severity on existing conditions, the alarm will need to be cleared and re-added.

Changing the alarm description of external relay alarm

The default alarm description for an external relay alarm is `Alarm on External Relay relayID ($$zhoneExternalRelayName$$)`.

You can customize individual MTAC external relay alarm descriptions in the `UDAlarmDescription.properties` file.

- To change the alarm description for the external relay alarm on the MALC with IP address 100.100.100.100, enters:

```
EXTERNAL_RELAY_2_ALARM.100.100.100.100=MTAC external alarm
```

- To change the alarm description for all external relay alarms with Relay ID 2, enters:

```
EXTERNAL_RELAY_2_ALARM=MTAC external alarm
```

All entries that specify an IP address take precedence over entries without an IP address. For example, if both the following entries exist:

```
EXTERNAL_RELAY_3_ALARM=MTAC
```

```
EXTERNAL_RELAY_3_ALARM.100.100.100.100=ALARM
```

A relay 3 alarm on the 100.100.100.100 device will show as ALARM. And a relay 3 alarm on all other devices will show as MTAC.

ONU/ONT serial number format (Hexa-decimal, Decimal)

DZMS supports one type of ONU/ONT serial number format displayed or set at a time. It could be either decimal or hexa-decimal.

To switch the ONU/ONT serial number format between decimal and hexa-decimal, change the **UseOnuFsanSN** flag in *NetHorizon* and *CS* properties using DZMS Web Client and NH Client. If this flag is true (this is the default value), all ONT serial numbers will be displayed and set in hexa-decimal format. If this flag is false, all ONT serial numbers will be

displayed and set in decimal format. After that, DZMS client has to log back in to make the changes take effect.

Display ONT/ONU MAC address

By default, the ONT MAC address field is not displayed in the **View/Modify GponOnu Physical Configuration** dialog.

The ONT MAC address field can be displayed in the DZMS client by changing the **ShowOntMacAddress** flag to true in the *NetHorizon* properties. By default, this value is set to false. After this change, the DZMS client has to log back in to make the changes take effect.

The ONT MAC address field will display the MAC address of the connected ONTs. The calculations of the MAC addresses are different for the ONT types. In case the ONT MAC Address field reported wrong MAC address, you can always force it matches to the ONT types by selecting the ONT type in the Device Type field in the **Modify GponOnu Physical Configuration** dialog. You can find the correct MAC address and ONT type in the CPE WEB UI.



Note: Make sure you selected correct ONT models in the Device Type field. The Device Type field will force the ONT MAC Address field to change according to your selection.

By setting **ShowOntMacAddress** flag to true in the *NetHorizon* properties, the Device Type field will also be shown in the **View/Modify GponOnu Physical Configuration** dialog. The Device Type field has three radio buttons: 420x/421x, 24xx/26xx/422x/424x/9xxx, and Other. If users selected Device Type as 420x/421x, ONU MAC address is increased by 1. If users selected Device Type as other, ONU MAC address will not be calculated. After selected Device Type, click **Modify** to make the changes take effect.

Enable wildcard searches

Performing wildcard searches in the **Find Objects** dialog is prohibited by default.



Caution: If a wild card search returns too many results, it could force a DZMS restart.

If users want to enable this feature, change the **restrictSearchsInFindObjectWindow** flag to false in the *NetHorizon* properties. By default, this value is set to true. After this change, the DZMS client has to log back in to make the changes take effect. To access the **Find Objects** dialog, select **Tools> Find Objects** menu item from the menu bar.

Hide the Find Object menu item

The **Tools> Find Objects** menu item could be hidden in the menu bar. To hide this menu item, change the **showFindObject** flag to false in the *NetHorizon* properties. By default, this value is set to true. After this change, the NetHorizon client has to log back in to make the changes take effect.

RADIUS client support in DZMS

Remote Access Dial In User Service (RADIUS) is used in a wide variety of places, from cellular network providers having millions of users to a small WISP start-up providing the local neighborhood with Internet connectivity to enterprise networks that implement Network Access Control (NAC) using 802.1x to ring fence their network. RADIUS is found in all these places and more. RADIUS server offers the implementation of Authentication, Authorization, and Accounting (AAA) services.

In order to take the advantage AAA layer provided by general RADIUS servers, DZMS server now supports the logins authenticated using RADIUS servers.



Note: DZMS server does not use the authorization and accounting features provided by RADIUS server at this point. Current DZMS does its own authorization and accounting.

To enable the support for RADIUS client in DZMS, users must perform the following procedure.

Enabling RADIUS client support in DZMS

- 1 Configure RADIUS server to work with DZMS users, use the following procedure.
 - a For procedures on RADIUS server configuration, please refer to your RADIUS server configuration document. This document assumes that users have already configured RADIUS server.

- b** Configure users in RADIUS server to contain DZMS login user name and password. Without such configuration, enabling Radius in DZMS server will reject DZMS logins as DZMS will encounter RADIUS authentication failures.



Note: RADIUS server must be configured with all valid DZMS admin and non-admin users and passwords by modifying */etc/freeradius/users*. This file location might be different if a different RADIUS server than FreeRadius server is used.

- 2** Enable RADIUS on the DZMS server, use DZMS Web Client or NH Client.

Table 10: RADIUS service (Radius properties) variables

Variable	Default value	Description
AccountingPort	1813	Accounting port in the range of 1025 to 65535.
AuthorizationPort	1812	Authentication port in the range of 1025 to 65535.
EnableRadius	false	Setting to true/false will enable / disable RADIUS server check during DZMS/OSS client login.
EncryptionProtocol	CHAP	Based on highest authentication level supported by RADIUS server, this flag can be set accordingly for using in RADIUS client logins done by DZMS server, make sure that RADIUS server supports the protocol that is set: CHAP PAP MSCHAPv1 MSCHAPv2
RadiusServer	Hostname.domainname.com	Host Name of Radius Server.
SharedSecret	testing123-2	Shared secret used for authentication between RADIUS Server and Clients.
TimeOut	1000	Represents the waiting time of RADIUS client before declaring response as non-reachable. The value is RADIUS server dependent. The unit is milli seconds.

- 3** Now, when users logging in to the DZMS client, must match the User Name and Password that set in the RADIUS server.



Note:

- If a DZMS user login credentials are validated by RADIUS server but there is no user found in DZMS database, login will still fail. DZMS user should first login as admin user and add a DZMS account for a new user.
 - DZMS server will provide the read/write permissions for specific operations based on user settings configured at the time of DZMS user addition in DZMS client. It will not use any authorization control list configured in RADIUS server.
 - Adding a DZMS user in DZMS database will not add the corresponding user in RADIUS server.
 - By the same token, deleting a DZMS user from DZMS database will not delete the user configured in RADIUS server.
 - DZMS server does not mandate the use of specific RADIUS server. Current code has been tested with FreeRadius RADIUS server.
-

LDAP client support in DZMS

Light Directory Access Protocol (LDAP) is an open, vendor-neutral, industry standard application protocol for accessing and maintaining distributed directory information services over an Internet Protocol (IP) network. LDAP server offers the implementation of Authentication, Authorization, and Accounting (AAA) services.

In order to take the advantage AAA layer provided by general LDAP servers, DZMS server now supports the logins authenticated using LDAP servers.



Note: DZMS server does not use the authorization and accounting features provided by LDAP server at this point. Current DZMS does its own authorization and accounting.

To enable the support for LDAP client in DZMS, users must perform the following procedure.

Enabling LDAP client support in DZMS

- 1 Configure LDAP server to work with DZMS users, use the following procedure.
 - a For procedures on LDAP server configuration, please refer to your LDAP server configuration document. This document assumes that users have already configured LDAP server.
 - b Configure users in LDAP server to contain DZMS login user name and password. Without such configuration, enabling LDAP in DZMS server will reject DZMS logins as DZMS will encounter LDAP authentication failures.



Note: LDAP server must be configured with all valid DZMS admin and non-admin users and passwords.

- 2 Enable LDAP on the DZMS server, use DZMS Web Client or NH Client.
- 3 Now, when users logging in to the DZMS client, must match the User Name and Password that set in the LDAP server.

Table 11: LDAP service (LDAP properties) variables

Variable	Default value	Description
EnableLdap	false	Setting to true/false will enable / disable LDAP server check during DZMS/OSS client login.
LDAPServer	ldap://hostname:389	Host Name of LDAP Server.
SecurityPL	"cn="+userName+", "+securityPL	LDAP Security Principal.
TimeOut	1000	Represents the waiting time of LDAP client before declaring response as non-reachable. The value is LDAP server dependent. The unit is milliseconds.



Note:

- If a DZMS user login credentials are validated by RADIUS server but there is no user found in DZMS database, login will still fail. DZMS user should first login as admin user and add a DZMS account for a new user.
- DZMS server will provide the read/write permissions for specific operations based on user settings configured at the time of DZMS user addition in DZMS client. It will not use any authorization control list configured in RADIUS server.
- Adding a DZMS user in DZMS database will not add the corresponding user in RADIUS server.
- By the same token, deleting a DZMS user from DZMS database will not delete the user configured in RADIUS server.
- DZMS server does not mandate the use of specific RADIUS server. Current code has been tested with FreeRadius RADIUS server.

AD client support in DZMS

Active Directory (AD) is Microsoft's proprietary directory service. It runs on Windows Server and allows administrators to manage permissions and access to network resources. The server offers the implementation of Authentication, Authorization, and Accounting (AAA) services.

In order to take the advantage AAA layer provided by Microsoft's AD servers, DZMS server now supports the logins authenticated using AD servers.



Note: DZMS server does not use the authorization and accounting features provided by AD server at this point. Current DZMS does its own authorization and accounting.

To enable the support for AD client in DZMS, users must perform the following procedure.

Enabling AD client support in DZMS

- 1 Configure AD server to work with DZMS users, use the following procedure.
 - a For procedures on AD server configuration, please refer to your AD server configuration document. This document assumes that users have already configured AD server.
 - b Configure users in AD server to contain DZMS login user name and password. Without such configuration, enabling AD in DZMS server will reject DZMS logins as DZMS will encounter AD authentication failures.



Note: AD server must be configured with all valid DZMS admin and non-admin users and passwords.

- 2 Enable AD on the DZMS server, use DZMS Web Client or NH Client.
- 3 Now, when users logging in to the DZMS client, must match the User Name and Password that set in the AD server.

Table 12: AD service (AD properties) variables

Variable	Default value	Description
ADServer	ldap://hostname:389	Host Name of AD Server.
DnsDomainSearchBase	DC=oak,DC=zhone, DC=com	LDAP domain search base. This value should not be changed or this value should only be changed when DNS domain changed with different while doing LDAP-AD SAMBA4 setup.
EnableAD	false	Setting to true/false will enable/disable AD server check during DZMS/OSS client login.
NetBiosDomainSP	oak	LDAP/AD Security Principal.
TimeOut	1000	Represents the waiting time of AD client before declaring response as non-reachable. The value is AD server dependent. The unit is milliseconds.

**Note:**

- If a DZMS user login credentials are validated by AD server but there is no user found in DZMS database, login will still fail. DZMS user should first login as admin user and add a DZMS account for a new user.
- DZMS server will provide the read/write permissions for specific operations based on user settings configured at the time of DZMS user addition in DZMS client. It will not use any authorization control list configured in AD server.
- Adding a DZMS user in DZMS database will not add the corresponding user in AD server.
- By the same token, deleting a DZMS user from DZMS database will not delete the user configured in AD server.
- DZMS server does not mandate the use of specific AD server.

Trap Multi-Threading

DZMS can now split trap processing into multiple threads, thereby increasing trap processing throughput.

In addition, when **numberOfTrapProcessingThreads** is set greater than 0, DZMS will create an individual rules engine per device being managed by DZMS. Each rules engine consumes approximately 1 MB of Java Heap Memory. For customers with a large number of devices being managed, they need to verify that enough memory is allocated for the Java Runtime Engine (JRE) before enabling this feature. Memory settings for the DZMS JRE are located in `/weblogic/startZMS.sh`.

To enable this feature, in the *FaultService* property, set **numberOfTrapProcessingThreads** to some number greater than 0. Recommended setting is 10.

OSS Returns “ProfileType_NOT_FOUND_ERROR” If No Records Found

The **showOssNoProfErrorOnZeroRecords** flag in the *CS* property shows `ProfileType_NOT_FOUND_ERROR` for the get and list operations that return 0 records.

To enable this feature, you need to set **showOssNoProfErrorOnZeroRecord=true** in the *CS* property.

Allow Downlink Bridges

In order to allow customers to continue to add bridges using the downlink bridge type, a backward compatibility mode was added to DZMS. This mode

is enabled by setting **allowDownlinkBridges=true** in the *NetHorizon* property.

When **allowDownlinkBridges** is true, the downlink bridgetype is displayed in the **bridgeType** comboBox. Otherwise, the downlink bridge type is not displayed and users must select the precise type of downlink bridge they are adding. ie., downlink-video, downlink-voice, ...

Even if **allowDownlinkBridges** is false, the downlink bridge type is still displayed for Malcs and older MXKs.

Configure Alpha LPS-36

To configure Alpha LPS-36, use the following procedure:

1. Log into the LPS-36 Web Interface and add a trap destination to DZMS.

Under Communications->Event Notification Destinations, select Add New Destination.

2. Update NWS property and include the ALPSMibs in mibsToLoad:

mibsToLoad=commonMibs SLMSMibs SkyZhoneMibs ALPSMibs

3. Update alarm rules file

In /zms/ZMS, concatenate non_slms_rules.ilr to rules.ilr

```
cat rules.ilr non_slms_rules.ilr > rules.ilr.temp
```

```
cp rules.ilr rules.ilr.original
```

```
mv rules.ilr.temp rules.ilr
```

4. restart DZMS

Configure Eltek Smartpack Alarm Reporting

To configure Eltek Smartpack alarm reporting, use the following procedure:

1. Configure Eltek device via its web interface:

Web into Eltek device and go to:

System Config->Device Settings->SNMP Settings

- a. Make sure "Web-based configuration" is selected. If not, select it.

In SNMP Settings page, enable the following:

"Send Off Traps"

"Trap Repeat Rate(minutes)" - this is used for alarm refreshing, where traps are re-sent to the trap destinations (like DZMS). Set to a number of minutes greater than 1, like 60.

Then save your changes by selecting the bottom left corner save icon which looks like a floppy disk.

- b. IMPORTANT: Enable "SNMP-based configuration" - this will cause device to reboot
2. Update NWS property and include the EltekSP2Mibs in mibsToLoad:
mibsToLoad=commonMibs SLMSMibs SkyZhoneMibs EltekSP2Mibs
In /zms/ZMS, concatenate non_slms_rules.ilr to rules.ilr
3. Update alarm rules
cat rules.ilr non_slms_rules.ilr > rules.ilr.temp
cp rules.ilr rules.ilr.original
mv rules.ilr.temp rules.ilr
4. restart DZMS

3

ADMINISTERING OPERATOR ACCOUNTS

This chapter describes how to use DZMS clients to set up operator accounts and how to assign operator permissions. It includes the following sections:

- [Overview, page 59](#)
- [Administering groups and operators as the admin user, page 67](#)
- [Configuring security policy as the admin user, page 72](#)
- [Administering operators as the user administrator, page 75](#)
- [Account information for operators, page 78](#)
- [About activating operator accounts, page 79](#)

Overview

Before you set up operators and assign permissions, you should understand how they are defined and how they function.

How objects are grouped

Operators and objects are associated with particular groups for the purpose of access control.

When operators are added, they are added to a particular group:

- The *admin* user can add an operator to any group.
- A user administrator can add an operator only to the user administrator's group.

Each operator can be assigned to only one group. An operator can manage objects only in his or her own group.

When objects are added to the network, they inherit the group ID of the operator performing the add operation:

- Once a region is added, all child objects added to the region (devices, cards, ports, and so on) inherit the same group ID.
- Once a customer is added, all child objects added to the customer (subscriber, voice gateways, and so on) inherit the same group ID.

- Once a subscriber is added, all child objects added to the subscriber (voice gateways, and so on) inherit the same group ID.

All operators and objects that were added before any groups were explicitly added inherit the default group ID.

How permissions function

Operators are assigned operator permissions. The DZMS administration manager uses these permissions to control each operator account. When the operator logs into a DZMS client, DZMS grants the operator access to specific types of objects and controls the actions the operator can perform on those objects:

- Only those objects accessible to an operator are visible in the DZMS client.
- Only those management functions granted to the operator are visible in menu selections in the DZMS client.
- Permissions can be granted to the operator are different in NetHorizon Client and DZMS Web Client. Because of that, the operator you created in NetHorizon Client can be displayed in DZMS Web Client, but only the name and the password info can be carried over not the permissions, and vice versa.

Permissions in NetHorizon Client

The permissions you can assign operators in NetHorizon Client are:

User Administration—Specifies whether the operator is a user administrator who can add other users or is a basic operator. Only the *admin* user can grant this permission.

OSS Gateway User—Specifies whether this is an operator that operates through OSS gateway.

Region Configuration—Specifies whether the operator can add, modify, and delete regions in the user's group or can only view configuration settings for regions in the user's group.

Equipment Provisioning—Specifies whether the operator can add, modify, and delete equipment (devices, cards, physical ports, and so on) in the user's group or can only view configuration settings and performance data for equipment in the user's group.



Note: DZMS users with Equipment Provisioning permission set to Read, can run MTAC, SELT and DELT tests as long as Diagnostics Administration permission is also enabled. DZMS users with Equipment Provisioning permission set to Write, continue to be able to run the MTAC, SELT and DELT tests by default.



Note: DZMS users with Equipment Provisioning permission set to Read can now also perform an OMCI ONT Reboot as long as Diagnostics Administration permission is also enabled.

DZMS users with Equipment Provisioning permission set to Read but with Diagnostics Administration permission disabled are not able to perform an OMCI ONTs Reboot.



Note: DZMS users with Equipment Provisioning permission set to Write can access to the **Change Administration Status** dialog on uplink and line card ports.

Subscriber Provisioning—Specifies whether the operator can add, modify and delete physicals and services on a line card in the user's group, and whether the operator can modify adsl-alarm and adsl-config profiles.

Port Admin Status Provisioning—Specifies whether the operator can view and modify the admin status of the ports. Operators with Port Admin Status Provisioning set to Write and Equipment Provisioning set to Read, can view the admin status of all ports, and change the admin status of the line card ports. If operators with both Port Admin Status Provisioning and Equipment Provisioning set to Write, or if they with only Equipment Provisioning set to Write, they can view and change admin status of all ports.

To change the admin status of the ports at port level, you can use the **Change Administration Status** dialog. To access this dialog, right-click a port in the tree view, then select Change Admin Status.

Bridge Admin Status Provisioning—Specifies whether the operator can modify the admin status of a bridge on a line card or not.

If the operators with the Bridge Admin Status Provisioning set to Write, they can change the admin status of the bridge on a line card. Otherwise, they cannot change the admin status.

Note that if the operators with the Subscriber Provisioning set to Write, no matter they set the Bridge Admin Status Provisioning to Write or Read, they can change the admin status of the bridge on a line card. Setting the Subscriber Provisioning to write is the equivalent of creating a super-user of the Bridge Admin Status Provisioning permission.

To change the admin status of the bridge on a line card, you can use the **Admin Up** and the **Admin Down** buttons in the **Manage Bridge Logicals** dialog. To access this dialog, right-click the device in the navigation tree, then select **Manage Bridging**.

ONT Provisioning—Specifies whether the operator can provision ONTs.

Note that the GPON ONT Provisioning permission type was added to Operator Configuration but is not visible by default. To enable this permission type set "showGponOntPermission" to "true" in *NetHorizon.properties*. Changing this property file requires re-signing of the jar file. Please call DZS Quality & Service for instructions.

ONT Reboot—Specifies whether the operator can reboot ONTs or not.

If the operators with the ONT Reboot set to Write, they can reboot ONTs. Otherwise, they cannot reboot ONTs.

Note that if the operators with the Equipment Provisioning set to Write, no matter they set the ONT Reboot to Write or Read, they can execute ONT reboot. Setting the Equipment Provisioning to write is the equivalent of creating a super-user of the ONT Reboot permission.

ONT Resync—Specifies whether the operator can resync ONTs or not.

If the operators with the ONT Reboot set to Write, they can resync ONTs. Otherwise, they cannot resync ONTs.

Note that if the operators with the Equipment Provisioning set to Write, no matter they set the ONT Resync to Write or Read, they can execute ONT resync. Setting the Equipment Provisioning to write is the equivalent of creating a super-user of the ONT Resync permission.

Trap/Alarms Administration—Specifies whether the operator can view traps and alarms and acknowledge, assign to troubleshooter, and close alarms of objects in the user's group or can only view traps and alarms of objects in the user's group.

User Idle Session Auto Logout—Specify whether the idle session auto logout feature is applied to the operator. If this parameter is disabled, the NetHorizon client session will never auto logout. For instance, an operator may be allowed to leave his/her session idled for indefinitely length of time while viewing alarm viewer or performance monitor chart.



Note: The *admin* user is a special account whose permissions cannot be changed. The *admin* user can only manage groups and users and grant permissions. And the admin user is the only user that can configure DZMS security policy.

Example user privilege setting case in NetHorizon

If you create a user with the below privilege:

- **OSS Gateway User** = No
- **Subscriber Provisioning** = Write
- **Port Admin Status Provisioning** = Write
- **All other options** are Disabled and Read.

Then, you will be able to do the following:

- Add/delete/modify pots->voip and isdn->voip connections.
- Add/delete/modify bond groups, except for the alarm properties.
- Add/delete/modify bridges/IP interfaces/IP hosts on line cards.
- Change admin status on line card physicals and bond groups.
- View everything.

Permissions in DZMS Web Client



Note: DZMS Web Client provisioning is currently limited to Unified Service Provisioning for GPON ONTs and Ethernet Bridge provisioning.

DZMS Web Client can be launched from NetHorizon Client Tools->Launch Web Client or accessed directly from the web browser using `http://zmsIPaddress:7001` in ZMS version 3.3 and earlier, `http://zmsIPaddress:8080` for version 3.4 and later, and `http://dzmsIPaddress:8080` for DZMS 4.x and later. When initially launching the DZMS Web Client, log in as "admin" user using same password as you do for NetHorizon Client. All NetHorizon client users will have an account for the DZMS Web Client. However, by default they all only have the "Default" Group which initially does not have any permissions. To quickly give all users some permissions, edit the "Default" group and add permissions for that group and save your changes. More advanced Groups can be created and assigned to users as is desired by the admin user.

By default, there are eight Permission Groups. Each Permission Group has its default selected permissions. You can create new Permission Groups and modify the permissions for each group as you desired.

The default permission groups and default permissions of the selected groups in DZMS Web Client are shown in [Table 13](#):

Table 13: Default Permission Groups and Default Selected Permissions in DZMS Web Client

Default Permission Groups	Default Permissions of the Selected Group	Description of Permissions
Administration	ADMINISTRATION APP	Specifies whether the operator can access the ADMINISTRATION Menu and use the Administration application. Administration group is the default group for the admin user.
Default	None	Default group is created for all new users created from NetHorizon.

Table 13: Default Permission Groups and Default Selected Permissions in DZMS Web Client

Default Permission Groups	Default Permissions of the Selected Group	Description of Permissions
Fault	FAULT APP	Specifies whether the operator can access the FAULT Menu and use the Fault application.
	MANAGE ALARMS	Specifies whether the operator can acknowledge, modify, and delete alarms.
Network Elements	NETWORK ELEMENTS APP	Specifies whether the operator can access the NETWORK ELEMENTS Menu and use the Network elements application.
	DEVICE ADD	Specifies whether the operator can add devices.
	DEVICE DELETE	Specifies whether the operator can delete devices.
	CARD ADD	Specifies whether the operator can add cards.
	CARD DELETE	Specifies whether the operator can delete cards.
Reports	REPORTS APP	Specifies whether the operator can access the REPORTS Menu and use the Reports application.
Service Provisioning	SERVICES APP	Specifies whether the operator can access the SERVICES Menu and use the Services application.
	SERVICE PROVISIONING	Specifies whether the operator can have the permission to provision service(s).
	SERVICE UNPROVISIONING	Specifies whether the operator can have the permission to delete service(s).
	ASSIGN SERIAL NUMBERS	Specifies whether the operator can have the permission to Assign/Unassign GPON ONU Serial Numbers.
	ASSIGN REG ID	Specifies whether the operator can have the permission to Assign/Unassign Registration ID.
	SERVICE PROVISION MODIFY	Specifies whether the operator can have the permission to modify all profiles in Manage Services.

Table 13: Default Permission Groups and Default Selected Permissions in DZMS Web Client

Default Permission Groups	Default Permissions of the Selected Group	Description of Permissions
Templates	TEMPLATES APP	Specifies whether the operator can access the TEMPLATES Menu and use the Templates application.
	SERVICE TEMPLATE CREATE	Specifies whether the operator can create/modify service templates.
	SERVICE TEMPLATE DELETE	Specifies whether the operator can delete service templates.
	ADVANCED SERVICE CREATE	Specifies whether the operator can create/modify advanced service templates.
	SERVICE GROUP CREATE	Specifies whether the operator can create/modify service groups.
	SERVICE GROUP DELETE	Specifies whether the operator can delete service groups.
Super User	SERVICE APP	Specifies whether the operator can access the SERVICES Menu and use the Services application.
	FAULT APP	Specifies whether the operator can access the FAULT Menu and use the Fault application.
	NETWORK ELEMENTS APP	Specifies whether the operator can access the NETWORK ELEMENTS Menu and use the Network elements application.
	ADMINISTRATION APP	Specifies whether the operator can access the ADMINISTRATION Menu and use the Administration application. Administration group is the default group for the admin user.
	TEMPLATES APP	Specifies whether the operator can access the TEMPLATES Menu and use the Templates application.
	REPORTS APP	Specifies whether the operator can access the REPORTS Menu and use the Reports application.
	SERVICE PROVISIONING	Specifies whether the operator can have the permission to provision service(s).
	SERVICE UNPROVISIONING	Specifies whether the operator can have the permission to delete service(s).

Recommended sequence for setup tasks

In order to administer operator accounts and permissions, you need to be logged in as a user with the appropriate permissions. In addition, you need to perform tasks in the correct order. The following procedures describe the recommended sequences for administering accounts in the following network scenarios:

- A new DZMS network with no objects
- A DZMS network with existing objects

Scenario 1: Administering accounts in a new DZMS network

Use this procedure to administer accounts in a DZMS network that does not yet contain objects.

- 1 To add group and users, log into a DZMS client as the *admin* user. (The default username is **admin**, the default password is **DZSone23!**).
- 2 For security purposes, change the password of the *admin* user to a private password. For instructions, see [Changing the admin password as the admin user on page 67](#).
- 3 Create groups to contains operators and the objects they can manage. For instructions, see [Creating a group as the admin user on page 67](#).
- 4 Create a user administrator for each group. For instructions, see [Creating an operator account as the admin user on page 68](#).
- 5 Add operators and assign permissions. For instructions, see [Creating an operator account as the admin user on page 68](#).
- 6 To add network objects, login into a DZMS client as an operator with the appropriate **Write** permissions.
- 7 Add network objects. The objects inherit that operator's group ID.
- 8 To add additional operators, login into a DZMS client as a user administrator.
- 9 Add operators and assign permissions. For instructions, see [Creating an operator account as the user administrator on page 75](#).

The operators inherit that user administrator's group ID. The operators can manage objects only in that group. If the operators have **View Objects in All Groups** permission, they can view objects in other groups.

Scenario 2: Administering accounts in an existing DZMS network

In the case of DZMS networks created before any groups were explicitly added, all existing operators and objects reside in the default group. In order to allow new operator accounts to view and manage these existing objects, you need to add the accounts to the default group. Use this procedure to add accounts to the default group.

- 1 To add operators, log into a DZMS client as a user administrator.
- 2 Add operators to the default group and assign permissions. For instructions, see [Creating an operator account as the user administrator on page 75](#).

The operators inherit the default group ID. The operators can manage objects only in the default group. If the operators have **View Objects in All Groups** permission, they can view objects that are in other groups.

Administering groups and operators as the admin user

When you log in as the *admin* user, the tree display lists only groups and operators. As the *admin* user, your permissions limit you to only managing groups and users and assigning permissions.

Changing the admin password as the admin user

When you log in as the *admin* user, for security reasons, you should change the password to a private password.

To change the admin password:

- 1 In NetHorizon, click **Tools > Change My Password** from the menu bar. NetHorizon displays the **Change Password** dialog box.

Or

In DZMS Web Client, click **Change password** in top right menu bar. DZMS Web Client displays the **Change Password for Admin** page.

- 2 Enter the current password in the **Current Password** box.
- 3 Follow the password rules displayed on the screen to enter a new password in the **New Password** and the **Confirm Password** boxes.



Note: The password rules are set in the **Modify Security Policy** dialog box. For the detail refer to [Setting the password rules on page 72](#).

- 4 Click **OK**.

Be sure to record the new password, so you can provide it to any individual who needs to log in as the *admin* user to add groups and add operators to various groups.

Creating a group as the admin user

Using NetHorizon to create groups as the admin user:

- 1 In the tree display, navigate to the list of groups.

- 2 Right-click the groups container and click **Add**.
NetHorizon displays the **Add Groups** dialog box.
- 3 In the **Name** parameter, enter a name for the group.
- 4 Click **Add**.

You can now add operator accounts to the group.

Using DZMS Web Client to create groups as the admin user:

User groups are added or edited when creating an operator account see section [Creating an operator account as the admin user on page 68](#) below.

Creating an operator account as the admin user

When you create an operator account, you specify the user name and password, and assign user permissions. Specifically, you associate the operator with a particular group and a set of permissions for the purpose of access control.

Once you activate the account, the operator can download, install, and run the NetHorizon client or login to the DZMS Web Client application. The operator permissions define what actions the operator can perform in DZMS client.

Using NetHorizon to create operators as the admin user:

- 1 In the tree display, navigate to the list of operators.
- 2 Right-click the operators container and click **Add**.
NetHorizon displays the **Add Operator Configuration** dialog box.
- 3 Complete the parameters in the **Add Operator Configuration** dialog box:
 - Name**—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)
 - Password**—Enter a unique password for the user.
 - Confirm Password**—Re-enter the password for the user.
 - Account Status**—To activate the operator, choose **Active** from the drop-down list. To deactivate the operator, choose **Inactive** from the drop-down list.
(Be sure to record the user name and password, so you can provide it to the operator.)
 - Permissions**—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).
- 4 Click **Add**.

The operator inherits the user administrator's group ID. The operator can manage objects only in that group. If the operator has **View Objects in All Groups** permission, the operator can view objects in other groups.

Using Web Client to create operators as the admin user:

- 1 Click on Administration either in the Home screen menus or the main navigation menus.
- 2 Click **Create New User** button on far right of the **Users and Groups** panel and **Users** sub-panel
- 3 Complete the parameters in the **Create a New User** page:

Name—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)

Display Name—(optional) displays alternate name for Welcome message.

User Group—Either choose from drop down selection or click + to add or edit groups.

Email Address— (optional)

Account Status—To activate the operator, choose Active from the drop-down list. To deactivate the operator, choose Inactive from the drop-down list.

(Be sure to record the user name, so you can provide it to the operator. At this point the default login password will be dzs123 which the operator will be forced to update on first signin)

OSS Gateway User— gives access via this account to any installed DZMS OSS gateway API (optional feature)

Permissions—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).

- 4 Click **Save**.

Viewing an operator account as the admin user

Using NetHorizon to view an operator account as the admin user:

- 1 In the tree display, navigate to the list of operators.
- 2 Select the operator you want to view and click **Edit > View**.

NetHorizon displays the **View Operator Configuration** dialog box with the following parameters:

Name—Operator's user name.

Account Status—Account status for the operator (**Active** or **Inactive**).

Permissions—View permissions for the user. For information about permissions see [How permissions function on page 60](#).

- 3 When you have finished viewing the operator settings, click **Close**.

Using Web Client to view an operator account as the admin user:

To use DZMS Web Client to view an operator account:

- 1 Click on Administration in either the Home screen menus or the main navigation menus.
- 2 Click Users and Groups panel and Users sub-panel.
- 3 Click any user row to display the details of the account.

Modifying an operator account as the admin user

Using NetHorizon to modify an operator account as the admin user:

- 1 In the tree display, navigate to the list of operators.
- 2 Right-click the operator and click **Modify**.
NetHorizon displays the **Modify Operator Configuration** dialog box.
- 3 Make the necessary changes to the values in the **Modify Operator Configuration** dialog box:

Name—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)

Password—Enter a new password for the user.

Confirm Password—Re-enter the new password for the user. (Be sure to record the new password, so you can provide it to the operator.)

Account Status—To activate the operator, choose **Active** from the drop-down list. To deactivate the operator, choose **Inactive** from the drop-down list.

Permissions—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).

- 4 When you have finished modifying the operator settings, click **Modify**.

Using Web Client to modify an operator account as the admin user:

To use Web Client to modify an operator account:

- 1 Click on Administration either in the Home screen menus or the main navigation menus.
- 2 Click **Users and Groups** panel and **Users** sub-panel

- 3 Click any user name to edit the details of the account.
DZMS Web Client displays the **Settings for Users:** panel.
- 4 Make the necessary changes to the values in the **Settings for User:** panel:

Name—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)

Display Name—(optional) Used for Welcome display in navigation banner.

Email Address— (optional)

Account Status—To activate the operator, choose **Active** from the drop-down list. To deactivate the operator, choose **Inactive** from the drop-down list.

OSS Gateway User— gives access via this account to any installed DZMS OSS gateway API (optional feature).
- 5 You can change **Password** or **Membership Groups** using the named sub-panels.

Membership Groups—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).
- 6 When you have finished modifying the operator settings, click **Save**.

When modifications take effect

If you change a user name or password, the operator must supply the new login information the next time the operator logs in to a DZMS client.

If you deactivate the operator, the operator can no longer log in to a DZMS client.

Other changes you make to an operator account take effect the next time the operator logs in. If operators are running a DZMS client when you modify their accounts, let them know that they must exit and log in to their DZMS client again for the change to take effect.

Deleting an operator account as the admin user

If you want to permanently remove an operator from the system, you can delete the account. If you want to temporarily remove the operator from the system, you can deactivate the account (For instructions, see [Modifying an operator account as the admin user on page 70](#)).

Using NetHorizon to permanently delete an operator account as the admin user:

- 1 In the tree display, navigate to the list of operators.
- 2 Right-click the operator and click **Delete**.
- 3 At the confirmation prompt, click **OK**.

Using DZMS Web Client to permanently delete an operator account as the admin user:

- 1 In the home page, click **ADMINISTRATION** menu.
DZMS Web Client displays the **User** page, listing existing users in all groups.
- 2 Highlight the operator and click **Delete User** button.
- 3 At the confirmation prompt, click **Yes**.

Once you delete an operator, the user can no longer log in to any DZMS client.

Configuring security policy as the admin user

The admin user can define security policies that will be applied when user login. The security policies include password rules, password blacklist, user login failure handling, user idle session auto logout, and user IP address filter.

Setting the password rules

Set the password rules. Those password rules will be displayed when user change password.

- 1 In NetHorizon, click Tools > Configure Security Policy from the menu bar. NetHorizon displays the **Modify Security Policy Configuration** dialog box.

Or

In DZMS Web Client, click on Administration in either the Home screen menus or the main navigation menus. Then click the **Security Configuration** panel.

- 2 The **Password Rules** window displays the current password rule settings. Make the necessary changes to the values in the **Password Rules** window. Note that some fields are NetHorizon Client only.

Minimum Password Length — Set the minimum password length, within the range of 4 to 30 characters.

Maximum Password Length — Set the maximum password length, within the range of 4 to 30 characters, and could not be less than the value in the **Minimum Password Length**.

Must Have Special Characters — To include special characters in the password, click **Enabled**.

Must Have Mixed Upper and Lower Cases — To include mixed upper and lower cases characters in the password, click **Enabled**.

Must Have Mixed Alphabetic and Numeric — To include mixed alphabetic and numeric in the password, click **Enabled**.

Default Characters — By default, the valid characters for password are A-Z, a-z, and 0-9.

Default Special Characters — The default special characters are ! # \$ & ' () + , - . / : ; < = > ? @ [\] ^ _ ` { } ~ .

Select Special Character Set — If the **Must Have Special Characters** is enabled, then user can configure the special character set in this field.

- To include default special characters into the password, select the **Default Special Characters**.
- To include specific special characters into the password, select **Specific Special Characters**.

Input Special Characters — To create a set of specific special characters, input desired special characters in this field.

Do Not Allow Password Re-use — To prevent using old passwords, check **Enabled**.

Number of Old Password Stored(5..10) — Set the maximum number of old passwords to be stored, in the range of 5 to 10. The stored old passwords are the most recent passwords the user has used. Those stored old passwords could not be used now.

Password Expiration — Enable or Disable the password valid days setting.

Password Valid Days(30..90) — If password expiration is enabled, set the length of days a password is valid in this field. In the range of 30 to 90.

Days to Warn Before Expiration(5..15) — Set number of days before expiration that warning is given. In the range of 5 to 15 days.

- 3 In NetHorizon click **Modify**. In Web Client click **Save**.

The new password rules will appear in the **Change Password dialog** box.

Setting the advanced security policies

Set the advanced security policies.

- 1 For NetHorizon Client, click Tools > Modify Security Policy from the menu bar. NetHorizon displays the **Modify Security Policy Configuration** dialog box.

Or

For DZMS Web Client, click on Administration in either the Home Screen menus or the main navigation menus. Then click the **Security Configuration** panel.

- 2 Click the **Advanced** tab. Make the necessary changes to the values in the **Advanced** window: Note that some fields are NetHorizon Client only.

Password Blacklist — To block passwords listed in the password blacklist, check **Enabled**.

Password Blacklist — If Password Blacklist is **Enabled**, admin user can edit the password blacklist. Click **Edit** to open the **Password Blacklist Dialog**. A list of existing blocked passwords appears.

- To add more passwords into the password blacklist, click **Add** to open the **Blocked Password Dialog**. In the **Blocked Password Dialog**, input desired blocked password and click **OK**.
- To remove a password from the blacklist, select the password and click **Delete**.

Do Not Allow Username As Password — To allow password is same as username, uncheck **Enabled**. Not allow password is same as username, check **Enabled**.

User Login Failure Handle — To apply the setting on the allowed number of login attempts, check **Enabled**. To have no limitation on the login attempts, uncheck **Enabled**.

Allowed Number of Login Attempts — If **User Login Failure Handle** is **Enabled**, admin user can input number of login attempts allowed before locking user out in this field.

Generate ZMS Alarm on Login Failure — To generate an alarm on DZMS alarm viewer when login failed, check **Enabled**.

User Idle Session Auto Logout — To apply the idle session timeout settings, check **Enabled**.

Maximum Session Idle Time (minutes)— If **User Idle Session Auto Logout** is enabled, admin user can specify the length of idle time that triggers an auto logout. The unit is in minutes.

Time Before Logout Warning (minutes)— If **User Idle Session Auto Logout** is enabled, admin user can specify the length of time that a warning message appears before auto logout.

User IP Address Filter— To apply the IP address filter, check **Enabled**.

IP Address Whitelist— To define a list of IP address ranges which are valid for a user to login from, click **Edit** to open the **User IP Whitelist Dialog**. A list of existing valid IP address ranges appears.

- To add an IP address range into the IP address whitelist, click **Add** to open the **Add Client IP Address** dialog box. In the **Add Client IP Address** dialog box, input desired IP address range to both **Start IP address** and **End IP address** fields and click **OK**. (If it is a single IP address, input it to Start IP Address field.)
- To remove an IP address range or a single IP address from the IP address whitelist, select it and click **Delete**.

IP Address Blacklist— To define a list of IP address ranges which no user may login from, click **Edit** to open the **User IP Blacklist Dialog**.

- To add an IP address range into the IP address blacklist, click **Add** to open the **Add Client IP Address** dialog box. In the **Add Client IP Address** dialog box, input desired IP address range to both **Start IP address** and **End IP address** fields and click **OK**. (If it is a single IP address, input it to Start IP Address field.)
- To remove an IP address range or a single IP address from the IP address blacklist, select it and click **Delete**.

Maximum User Sessions— To define the maximum number of concurrent NetHorizon login sessions a single user may have, check **Enabled**, and then specify the **Maximum times the same operator can be logged in** field.

- 3 In NetHorizon click **Modify**, in DZMS Web Client click **Save**.

Administering operators as the user administrator

Creating an operator account as the user administrator

When you create an operator account, you specify the user name and password and assign user permissions.

Once you activate the account, the operator can download, install, and run the DZMS client application. The operator permissions define what actions the operator can perform in DZMS client.

Using NetHorizon to create operators as the user admin:

- 1 In the tree display, navigate to the list of operators.
- 2 Right-click the operators container and click **Add**.
NetHorizon displays the **Add Operator Configuration** dialog box.
- 3 Complete the parameters in the **Add Operator Configuration** dialog box:

Name—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)

Password—Enter a unique password for the user.

Confirm Password—Re-enter the password for the user.

Account Status—To activate the operator, choose **Active** from the drop-down list. To deactivate the operator, choose **Inactive** from the drop-down list.

(Be sure to record the user name and password, so you can provide it to the operator.)

Permissions—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).

4 Click Add.

The operator inherits the user administrator's group ID. The operator can manage objects only in that group. If the operator has **View Objects in All Groups** permission, the operator can view objects in other groups.

Using Web Client to create operators as the user admin:

It is same as [Using Web Client to create operators as the admin user: on page 69](#).

Viewing an operator account as the user administrator

Using NetHorizon to view an operator account as user admin:

- 1 If you are logged in to the NetHorizon as an operator with **User Administration** privileges, click **Tools > Configure Users**.

NetHorizon displays the **User Administration** dialog box, listing existing users in your group.

- 2 Right-click the operator you want to view and click **View**.

NetHorizon displays the **Operator Configuration** dialog box with the following parameters:

Name—Operator's user name.

Password—Asterisks in place of the operator's password.

Account Status—Account status for the operator (**Active** or **Inactive**).

Permissions—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).

Account Active Date and Time— View day(s) of week that account is activated, and time range per day when account is activated.

- 3 When you have finished viewing the operator settings, click **Close**.

Using Web Client to view operators as user admin:

It is same as [Using Web Client to view an operator account as the admin user: on page 70](#).

Modifying an operator account as the user administrator

Using NetHorizon to modify an operator account as the user admin:

- 1 If you are logged in to NetHorizon as an operator with **User Administration** privileges, click **Tools > Configure Users**.

NetHorizon displays the **User Administration** dialog box, listing existing users in your group.

- 2 Select the operator and click **Edit > Modify**.

NetHorizon displays the **Operator Configuration** dialog box.

- 3 Make the necessary changes to the values in the **Operator Configuration** dialog box:

Name—Enter the user name. Use only alphanumeric characters and the underscore (_) character. (Do not use blank characters.)

Password—Enter a new password for the user.

Confirm Password—Re-enter the new password for the user. (Be sure to record the new password, so you can provide it to the operator.)

Account Status—To activate the operator, choose **Active** from the drop-down list. To deactivate the operator, choose **Inactive** from the drop-down list.

Permissions—Assign permissions for the user. For information about permissions see [How permissions function on page 60](#).

Account Active Date and Time— Modify day(s) of week that account is activated, and set time range per day when account is activated.

- 4 When you have finished modifying the operator settings, click **Modify**.

Using Web Client to modify operators as the user admin:

It is same as [Using Web Client to modify an operator account as the admin user: on page 70](#).

When modifications take effect

If you change a user name or password, the operator must supply the new login information the next time the operator logs in to a DZMS client.

If you deactivate the operator, the operator can no longer log in to a DZMS client.

Other changes you make to an operator account take effect the next time the operator logs in. If operators are running a DZMS client when you modify their accounts, let them know that they must exit and log in to their DZMS client again for the change to take effect.

Deleting an operator account as the user administrator

If you want to permanently remove an operator from the system, you can delete the account. If you want to temporarily remove the operator from the system, you can deactivate the account (For instructions, see [Modifying an operator account as the user administrator on page 76](#)).

Using NetHorizon to permanently delete an operator account as the user admin:

- 1 If you are logged in to NetHorizon as an operator with **User Administration** privileges, click **Tools > Configure Users**.
NetHorizon displays the **User Administration** dialog box, listing existing users in your group.
- 2 Right-click the operator and click **Delete**.
- 3 At the confirmation prompt, click **OK**.

Once you delete an operator, the user can no longer log in to any DZMS client.

Using DZMS Web Client to permanently delete an operator account as the user admin:

It is same as [Using DZMS Web Client to permanently delete an operator account as the admin user: on page 72](#).

Account information for operators

As you define operator accounts, be sure to record the user name and password information.

Supply operators with the following information so they can download, install, and run the NetHorizon java-based application.

- URL for the DZMS secure web server.
 1. The URL of the NetHorizon client has the following format:
http://server_hostname:server_port_number/zms
For ZMS 3.3 and earlier example:
http://grove:7001/zms
For ZMS 3.4 and later example:
http://grove:8080/zms
For DZMS 4.x and later example:
http://grove:8080/dzms
 2. The URL of the DZMS Web Client has the following format:
http://server_hostname:server_port_number
For ZMS 3.3 and earlier example:
http://grove:7001
For ZMS 3.4 and later example:
http://grove:8080
For DZMS 4.x and later example:

`http://grove:8080`

- DZMS client user name and password.

About activating operator accounts

Before operators can log in to a DZMS client, their accounts must be activated. To activate an operator account, modify the user account and set the **Account Status** parameter to **Active**.

After the **User Login Failure Handle** is enabled, and the **Allowed Number of Login Attempts** is specified in the **Modify Security Policy Configuration** dialog box. If an operator attempts to log in to a DZMS client certain consecutive times, but fails (because the operator supplies the wrong user name or password), the administration manager automatically deactivates the operator account. The operator cannot log in to any DZMS client until you reactivate the account.

4 DZMS DATABASE SERVER ADMINISTRATION

This chapter describes basic DZMS database server administration with Oracle Database and Maria Database. It includes:

- [Oracle Database Server Administration, page 81](#):
 - [DZMS Oracle database utilities, page 81](#)
 - [Starting, Stopping, and Checking the Oracle Database, page 84](#)
 - [Oracle Database monitoring, page 84](#)
 - [Oracle Database backup and restore, page 85](#)
 - [Common Oracle database administration procedures, page 89](#)
 - [Backing up Oracle database archive logs, page 89](#)
- [Maria Database Server Administration, page 90](#):
 - [Starting, Stopping, and Checking the Maria Database, page 90](#)
 - [Maria Database backup and restore, page 90](#)

Oracle Database Server Administration

DZMS Oracle database utilities

A typical DZMS installation will include the following scripts and cron jobs to monitor and maintain the database. The user is oracle or root.

At installation time, you can configure many of the parameters for these scripts. After the installation is complete, modify these scripts using the **crontab** command.

Table 14: Database utilities

Script	Location	Run as	Description
<i>cold-backup.sh</i>	<i>zms/bin</i>	<i>oracle</i>	Performs a manual Cold backup of the oracle database. DZMS must be down. The cold backup is not recommended for daily use, rather upon DZS Quality & Service request.
<i>data-collector.sh</i>	<i>zms/bin</i>	<i>oracle</i>	Collects system configuration information that can be analyzed by DZS Quality & Service. Run it and provide the output file to DZS Quality & Service for analysis. The location and output file naming format are the following: <i>/var/tmp/ZMS_collector/</i> <i>YYYYMMDD_HH24MI/</i> <i>Zhone_data_collector_YYYYMMDD_HH24MI.tar.gz.</i>
<i>dbmonitor.sh</i>	<i>zms/bin</i>	<i>oracle</i>	Monitors the health of DZMS database for errors. Works based on an error white list (<i>/opt/oracle/zms/bin/ora-msg-wlist.lst</i>) with errors that can be safely ignored. It can send out email whenever an error is detected. It also creates a log file with all errors trapped (<i>/var/tmp/dbmonitor.log</i>). This script is scheduled in oracle's crontab at install time. To send email change the following: 1. change <i>dbmonitor.sh</i> line # 74 "RECIPIENT=some-address@some-domain.com" to a real email address. 2. crontab entry from "<i>/opt/oracle/zms/bin/dbmonitor.sh OFF</i>" to "<i>/opt/oracle/zms/bin/dbmonitor.sh ON</i>"
<i>dual-node-config.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This scripts is used to change the DZMS configuration to point it to different database server. It is mainly used on a two server setup (1 DZMS, 1 database) when the DZMS database server name changes. The script will update the DZMS configuration files with the new database server name.

Table 14: Database utilities

Script	Location	Run as	Description
<i>hot-backup.sh</i>	<i>zms/bin</i>	<i>oracle</i>	A cron job that creates a hot backup of a DZMS database. A hot backup does not require any downtime. The hot backups will also flush out old archived logs. If these are not flushed they will consume all their storage quota and the system will stop. This script is scheduled in oracle's crontab at install time to run every day at 2:00am.
<i>ora_shutdown.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This script is used by the OS service framework to stop the oracle database and listener. DO NOT MODIFY THE SCRIPT IN ANY WAY EXCEPT WHEN INDICATED BY DZS Quality & Service.
<i>ora_startup.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This script is used by the OS service framework to start the oracle database and listener. DO NOT MODIFY THE SCRIPT IN ANY WAY EXCEPT WHEN INDICATED BY DZS Quality & Service.
<i>recovery-menu.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This script will restore a database hot backup. It will present you with different available options. Select the best option in your case and follow the screen instructions.
<i>restore-cold-backup.sh</i>	<i>zms/bin</i>	<i>oracle</i>	Restores the database from a manual Cold backup taken using <i>cold-backup.sh</i> script. The script will present users with the available backups can use. Users decide which one to use and follow the screen instructions to restore the cold backup or perform other operations.
<i>ssh-setup.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This script is used to set passwordless ssh connectivity. Use it when indicated by DZS Quality & Service.
<i>sshUserSetup.sh</i>	<i>zms/bin</i>	<i>root</i>	This script is used to set passwordless ssh connectivity. Use it when indicated by DZS Quality & Service.
<i>version.sh</i>	<i>zms/bin</i>	<i>oracle</i>	This script will check the DZMS database schema version.

Table 14: Database utilities

Script	Location	Run as	Description
<i>zms-fast-upgr-extract.sh</i>	<i>zms/bin</i>	<i>root</i>	This script will create a DZMS schema export file which can later be imported into a different database. This is the only supported method to port the DZMS schema between databases.
<i>zms-fast-upgr-import.sh</i>	<i>zms/bin</i>	<i>root</i>	This script will import a DZMS schema export. The script is destructive meaning that it will erase any DZMS information in the database where the DZMS schema is imported. Use carefully.

Starting, Stopping, and Checking the Oracle Database

To start, stop or check the Oracle DB status, use the following commands:

```
<sudo> systemctl [start|stop|status] dbora
```

Note that command text in square brackets means choose one of the options. e.g. `[start|stop|status]`. Command text in angle brackets means text is optional. e.g. `sudo` (without angle brackets) is optional if logged in as oracle user, it is needed if not logged in as root user.

Oracle Database monitoring

The database monitoring script is called *dbmonitor.sh* and it is located in */opt/oracle/zms/bin*.

As part of the DZMS database installation, *dbmonitor*, a database monitoring script, gets scheduled. This script monitors the following:

- Database status (up and down)
- TNS listener status (up and down)
- Error messages in the alert log file

If the database is down, the TNS Listener is down or if there is an error message in the alert log file (*/opt/oracle/app/oracle/diag/rdbms/zhonecs/ZhoneCS/trace/alert.log*) an alert will be mailed to the DBA pager email address specified during the installation.

The database monitoring script is triggered from the Oracle's account crontab:

```
00 02 * * * /opt/oracle/zms/bin/hot_backup.sh
05 00 * * * /opt/oracle/zms/bin/dbmonitor.sh OFF
```

To use the database monitoring script, requires an argument which is “ON” or “OFF”. This argument (ON or OFF) will enable or disable sending email notifications to an specific recipient.

Notifications are turned off by default. To enable the notification, modify the script and change the following line with a valid email address:

```
# This is the account that will receive all messages
RECIPIENT=some-address@some-domain.com
```

The script works with a white list where certain database events can be flagged to be ignored. Such list is in `/opt/oracle/zms/bin/ora-msg-wlist.lst`.

The white list format has two columns:

```
ignr ORA-test
```

The first column is to specify the action. Possible actions are “ignr” for ignore and “mail” to mail the error message to the address previously specified.

```
mail ORA-00600
```

The second column is the error code from the database alert log. To add additional error codes just edit the file and add the action and code.

All database events starting with an “ORA-” code will get registered in the dbmonitor log file (`/var/tmp/dbmonitor.log`). If notifications are turned off, it is recommended to check this file from time to time to make sure everything is working fine with the database.

Oracle Database backup and restore

This section describes the following types of backup.

- [Cold backup of the DZMS Oracle database on page 86](#)
- [Hot backup of the DZMS database on page 87](#)
- [Restoring the Oracle database on page 87](#)



Note: The database backup strategy calls to be able to restore up to 72 hours prior. To achieve that window RMAN, Oracle's backup and recovery tool, will keep all required backup pieces and archived logs. RMAN also is responsible of flushing all files that are not needed

anymore. This is yet another reason why the backups should be scheduled, otherwise the backup files will pile up and consume the free disk space and potentially stopping the database and DZMS system.

The 3-day window can be customized. If you decide to do that, please contact support on instructions on how to do it and consider that you will require more disk space to keep a longer backup history.

If you are required to move the backup files out of the server, you can copy all the files under `/opt/oracle/zms/backup/oraback`.

Cold backups are not recommended as they require DZMS and database to be stopped prior to creation.

When moving/copying a database from one server to another it is recommended to use export-import method via the

"zms-fast-upgr-extract.sh" and "zms-fast-upgr-import.sh" scripts. A cold or hot backup can be used for that purpose but given highly technical steps involved the ONLY supported method is the export-import

Cold backup of the DZMS Oracle database

Cold backup is not recommended for daily use. Only use cold backup upon DZS Quality & Service request.

The cold-backup.sh script shuts down the database and performs a full backup of the database files.

- 1 Log in to the DZMS database server (as *oracle*).
- 2 Issue the following commands to execute the cold backup script:

```
/opt/oracle/zms/bin/cold-backup.sh ZhoneCS
```



Note: It is recommended that you copy the *ZhoneCS_<date>.tar* file to another location in case you need to remove the DZMS installation.

- 3 When prompt, press **Y** to proceed the cold backup.
- 4 When prompt, press **Y** to proceed the cold backup.
- 5 When prompt, press **Y** to confirm and proceed the cold backup.
- 6 When the backup completes, the script restarts the DZMS database server. It then displays the following message and returns to the UNIX prompt:

```
Cold backup has completed...
```

You can find the backups at `/opt/oracle/zms/backup/oraback`

Hot backup of the DZMS database

You can schedule a Hot backup for your DZMS database after part of the installation process. You will find a *hot-backup.sh* job in the Oracle cron. You can change the frequency and time of this job by using the **crontab** command to modify Oracle cron.

Issue the following command as oracle to execute the hot backup script:

```
/opt/oracle/zms/bin/hot-backup.sh
```

When the backup completes, the following message appears:

```
Hot backup has completed...
```

During a Hot backup, check the Hot backup log file:

```
/opt/zms_logs/db-hot-bckp_<YYYYMMDD_HHMM>.log
```

You can find the backups at */opt/oracle/zms/backup/oraback*.

Restoring the Oracle database

There are two methods of restoring the Oracle database from a backup:

- [Restoring the Oracle database from a Cold backup on page 87](#)
- [Restoring the Oracle Database from a Hot backup on page 88](#)

Restoring the Oracle database from a Cold backup



Caution: Only qualified DBAs should perform an active cold restore. For more information, contact DZS Quality & Service.

You can restore the active or the standby database from a Cold backup that was made of the standby database. You might need to restore from an active cold backup if an upgrade fails and you need to revert it back to the active Cold backup taken prior to the upgrade. Restoring from a Cold backup will restore the database to the state it was in at the time of the last backup.

A database restore from a cold backup is done via */opt/oracle/zms/bin/restore-cold-backup.sh*. This script needs an argument: ZhoneCS.

- 1 Log in to DZMS database server (as Oracle).
- 2 Issue the following command to run the standby database administration script:

```
/opt/oracle/zms/bin/restore-cold-backup.sh  
ZhoneCSstdby
```

The warning message appears:

```
----- WARNING WARNING WARNING -----  
This script will restore a cold backup of the ZMS database(s)  
During the restore process the ZMS database(s) will not be accessible  
by ZMS clients. Depending on the size of your installation this  
restore can take from 5 minutes to 30 minutes.  
nOnce the restore is completed, the database will be as it was when the
```

```
backup was created. ALL INFORMATION ENTERED AFTER THAT TIME WILL BE LOST.  
Do you wish to proceed ? N|Y
```

3 Enter y.

4 When prompted enter a full name of the cold backup you want to restore.

For example:

```
/opt/oracle/zms/backup/oraback/  
cold_bckp_20170327_1704.tar.gz
```

The following message appears:

```
File you want to restore is : /opt/oracle/zms/backup/  
oraback/cold_bckp_20170327_1704.tar.gz
```

User confirmation required.

```
Is that the correct backup file? Y|N
```

5 Enter y. After confirmation, restore starts.

When the restore completes, the script restarts the DZMS database server.
It then display the following messages.

```
Confirmation received, restore will start.  
==> Please wait while database is being restored. Mon Mar  
27 17:18:18 PDT 2017.  
==> Restoring database : Mon Mar 27 17:18:18 PDT 2017.  
Please be patient.  
==> Database restore completed sucessfully : Mon Mar  
27 17:18:52 PDT 2017.  
==> Database has been restored . Mon Mar 27 17:18:52  
PDT 2017.  
==> Starting database.  
==> Database started.
```

Restoring the Oracle Database from a Hot backup

Each recovery scenario is different. DZS recommends that each installation site have a qualified DBA in their staff to perform a recovery using the Hot backups. Contact DZS Quality & Service for additional support.

Issue the following command to execute the Hot backup script as oracle:

```
/opt/oracle/zms/bin/recovery-menu.sh
```

The following menu appears. Select the option as you needed.

Please select from the following options:

- 1) Repair database.
(Let Oracle recover the database)
- 2) Restore database based on previous point in time.
(You must enter the desired time to restore)
- 3) Restore database based on a previous backup.

```

        (Choose a backup from a list of available
backups)
4) List available backups.
        (List backups which are available for restore)
X) to exit

```

Common Oracle database administration procedures

This section describes the following DZMS database procedures you might need to perform to configure and maintain your DZMS database:

Changing the administrator password

To log in as a DZMS database administrator, use the user name *zdba* and the password *zdba*. As the database administrator, you have full privileges to the DZMS database server.

It is highly recommended that you change the administrator password to a private password. Use the following procedure to do so:

- 1 Log in to the primary DZMS database server, using the user account *ZMS_ORACLE_USER*.
- 2 Issue the following command to change the password:

```

sqlplus /nolog
SQL>connect /as sysdba
SQL>alter user zdba identified by new_password;
exit

```

where *new_password* is the new private password for the user.

Be sure to record the new password, so you can provide it to any administrator who needs to upgrade or maintain the DZMS database server.

Backing up Oracle database archive logs

The DZMS system generates database archive logs, listing database activity. DZMS generates the archive logs to the path:
u01/oradata/FRA/ZHONECS/archivelog/<YYYY_MM_DD>/<Name>.arc.
 The logs are named dynamically and have the suffix *.arc*.

Be aware that these archive logs can quickly become large, especially during periods of high activity. If the files reach directory capacity, the DZMS database halts until more space becomes available.

Keep in mind that the archive logs are required in the recovery process. Do not delete the logs without saving them to permanent medium.

Maria Database Server Administration

Starting, Stopping, and Checking the Maria Database

To start, stop or check the MariaDBstatus, use the following commands:

```
<sudo> systemctl [start|stop|status] mariadb
```

Note that command text in square brackets means choose one of the options. e.g. `[start|stop|status]` . Command text in angle brackets means text is optional. e.g. `sudo` (without angle brackets) is optional if logged in as oracle user, it is needed if not logged in as root user.

Maria Database backup and restore

This section describes how to backup and restore MariaDB.

Backing up the MariaDB

A Maria database backup is done via `/opt/oracle/zms/bin/mdb_bckp.sh`.

- 1 Log in to DZMS database server (as Root or Oracle user).
- 2 Issue the following command to run the Maria database administration script:

```
/opt/oracle/zms/bin/mdb_bckp.sh
```

The following message appears:

```
==> Backing up ZhoneCS database: Fri Aug  2 09:47:04 PDT 2019. Please be
patient.
==> Completed ZhoneCS database backup. Fri Aug  2 09:47:06 PDT 2019
Backup files :
total 77M
-rw-r----- 1 root root 16K Aug  2 09:47 aria_log.00000001
-rw-r----- 1 root root  52 Aug  2 09:47 aria_log_control
-rw-r----- 1 root root 324 Aug  2 09:47 backup-my.cnf
-rw-r----- 1 root root 972 Aug  2 09:47 ib_buffer_pool
-rw-r----- 1 root root 76M Aug  2 09:47 ibdata1
-rw-r----- 1 root root 2.5K Aug  2 09:47 ib_logfile0
drwx----- 2 root root 4.0K Aug  2 09:47 mysql
drwx----- 2 root root  20 Aug  2 09:47 performance_schema
-rw-r----- 1 root root  79 Aug  2 09:47 xtrabackup_checkpoints
-rw-r----- 1 root root 467 Aug  2 09:47 xtrabackup_info
drwx----- 2 root root 60K Aug  2 09:47 zhonecs
```

Restoring the MariaDB from a backup

A Maria database restore is done via `/opt/oracle/zms/bin/mdb_restore.sh`.

- 1 Log in to DZMS database server (as Root or Oracle user).

- 2 Issue the following command to run the Maria database administration script:

```
/opt/oracle/zms/bin/mdb_restore.sh
```

The following message appears:

```
----- WARNING WARNING WARNING -----
This script will restore a database backup of the ZMS MariaDB database.
During the restore process the ZMS database will not be accessible by
any ZMS client. Depending on the size of your installation this restore
can take from just a couple minutes to 30 minutes or more.
Please be patient.
Once the restore is completed, the database will be as it was when the
backup was created. ALL INFORMATION ENTERED AFTER THAT TIME WILL BE LOST.
Do you wish to proceed ? N|Y
```

- 3 Press Y.

The following message appears:

```
Selected to PROCEED with database restore.
Available backups starting with the most recent :
(file format : yyyyymmdd_hhmi)
20190724_0926
20190802_0947
Please enter the full name of the backup you want to restore :
```

- 4 Enter the full name of the backup file you want to restore.

This example enters:

```
20190802_0947
```

The following message appears:

```
The backup you want to restore is : 20190802_0947
User confirmation required.
Is that the correct backup ? Y|N
```

- 5 Press Y.

The following message appears:

```
Confirmation received, restore will start.
* INFO : Database was sucessfully restored.
*****
IMPORTANT IMPORTANT IMPORTANT IMPORTANT IMPORTANT IMPORTANT
Database has been restored to how it was on 2019-08-02 at 09:47 hrs.
It is now open and ready to use. ZMS can be restarted.
All restore actions can be find here:
/opt/zms_logs/mdb-restore-20190802_0948.log
```


5

MONITORING DZMS LOGS

This chapter describes how to use DZMS logs. It includes the following information:

- [Alarm log, page 94](#)
- [Audit logs, page 94](#)
- [Task logs, page 96](#)
- [Debug log, page 97](#)
- [Forwarded alarm log, page 97](#)
- [Forwarded trap log, page 98](#)
- [Trap log, page 98](#)
- [Server log, page 98](#)
- [DZMS error log, page 101](#)

Overview

The DZMS application server creates the following logs:

- Alarm log (*yyyy_mmm_dd_alarm.log*)
- Audit log (*yyyy_mmm_dd_ZmsAudit.log*)
- Task log (*yyyy_mmm_dd_Task.log*)
- Debug log (*yyyy_mmm_dd_debug.log*)
- Forwarded alarm log (*yyyy_mmm_dd_AlarmForward.log*)
- Forwarded trap log (*yyyy_mmm_dd_TrapForward.log*)
- Trap log (*yyyy_mmm_dd_trap.log*)
- DZMS error log (*yyyy_mmm_dd_ZMSErrors.log*)
- Server log (*zms.log*)

Each of these logs (with the exception of the DZMS server log) is created daily and stored in the following location:

install_directory/opt/weblogic/yyyy_mmm_dd_LogName.log

where `yyyy_mmm_dd` is the timestamp for when the log was created. These log files are purged periodically based on the configuration of the *ADS.properties* file. The default is 31 days. For details, see [Modifying the information written to the audit log on page 95](#).

If you want to save the log file information, you should periodically backup the log files to another location.

Alarm log

The DZMS system generates an alarm log (*alarm.log*), listing all alarms generated on the network. The alarm log lists entries sequentially by date and time, with the most recent trap appearing at the end of the file.

The following sample illustrates an alarm log entry:

```
Tue Feb 05 18:08:58 EST 2002: Received LINK_DOWN_ALARM
Severity: Critical
ALARM: Communication link is about to enter the down
state
Device: NEDevice
Shelf: 1
Slot: 1
Port: 7

Tue Feb 05 18:10:41 EST 2002: Cleared LINK_DOWN_ALARM
Device: NEDevice
Shelf: 1
Slot: 1
Port: 7
---
```

Audit logs

DZMS generates an audit log, listing all operator activity. The log lists entries sequentially by date and time, with the most recent event appearing at the end of the file.

The DZMS system stores the audit log files in the following path:
install_directory/opt/weblogic/yyyy_mmm_dd_ZmsAudit.log.

Entries in the log file can be viewed from ZMS by selecting **Tools > View Audit Log**.

Audit log entries include the fields listed in [Table 15](#).

Table 15: Audit log fields

Audit log field	Description
Time	Date and time of the activity.
User	User performing an operation.

Table 15: Audit log fields

Audit log field	Description
Operation Type	Operation performed by DZMS. Possible values are: Add: An object has been added. Modify: An object has been modified. Delete: An object has been deleted. DR_List: An objects attributes have been listed. DR_Restore: A device configuration has been restored. Login: A user has logged in. Logout: A user has logged out.
Authorization	Whether the request was granted or denied.
Device Name	Name of affected device.
Object Type	The type of object affected.
Message	An explanatory message describing the event.
Operation Request	Operation requested by the user.

Modifying the information written to the audit log

The information that DZMS sends to the audit log can be configured in the *ADS* property. Note that since each level of auditing requires more resources (such as disk space, time, and CPU resources), some tuning may be required in order to achieve the desired outcome.

The following table describes the options in the *ADS* property. Note that changes to the log files take effect after the DZMS application server is restarted.

Table 16: ADS property options

Audit log field	Description
bulkStatsLogPurge	Indicates the number of days DZMS should store the bulk statistics logs before purging them. The default is 31.
gctimer	How often the memory optimizer runs to clean up any available memory within DZMS. This option should not be changed from its default of 60 minutes.

Table 16: ADS property options (Continued)

Audit log field	Description
logLogin	Whether attempts to log in should be logged in the audit file. The default is 1 (added to log).
logLogout	Whether attempts to log out should be logged in the audit file. The default is 1 (added to log).
logPurge	Indicates the number of days DZMS should store the logs before purging them. The default is 31. Must be greater than 1.
logRequest	Whether every operational request made to DZMS is logged. This option enables logging of the user, request type made, object type, and unique object ID for every request. This is the lowest form of request logging available that accurately depicts the interaction of clients with the DZMS server. The default is 1 (added to log).
logModDetails	Whether additional details of modification requests should be logged. The additional information includes every attribute that was modified and its prior and new value. Note that this option can substantially add to the size of the log files. The default is 0 (not added to log).
logAddDetails	This option logs every attribute of an object whenever it is added to DZMS. These attributes are unique to each object, and include information such as name, index, admin status, and so on. Note that this option can substantially add to the size of the log files. The default is 0 (not added to log).

Task logs

DZMS generates a task log, listing all activity for these four tasks: Auto Discovery, Device Backup, ConfigSync, and Download Image. The log lists entries sequentially by date and time, with the most recent event appearing at the end of the file.

The DZMS system stores the task log files in the following path:
install_directory/opt/weblogic/yyyy_mmm_dd_Task.log.

Entries in the log file can be viewed from ZMS by selecting **Tools > View Task Log**.

Task log entries include the fields listed in [Table 17](#).

Table 17: Task log fields

Task log field	Description
Time	Date and time of the activity.
Task Type	The type of task: ALL, Auto Discovery, Device Backup, ConfigSync, Download Image.
Message Type	Whether the task was started or finished.
Status	Whether the task was succeeded or failed.
Description	A brief description describing the events.
Object Type	The type of object affected.
Message	An explanatory message describing the event.
Operation Request	Operation requested by the user.

Debug log

The DZMS system generates a debug log (`debug.log`), that stores a variety of debugging information related to the DZMS server operations. This log file is typically only useful for DZS development engineers.

The DZMS system generates the debug log to the path: `install_directory/opt/weblogic/yyyy_mmm_dd_debug.log`.

Forwarded alarm log

The DZMS system generates a forwarded alarm log (`AlarmForward.log`), listing information about alarms that are forwarded to IP addresses on the network. The forwarded alarm log lists entries sequentially by date and time, with the most recent alarm appearing at the end of the file.

The following sample illustrates an alarm log entry:

```
Wed Jan 30 16:46:18 EDT 2002: Forwarded
CARD_PROVISIONING_FAILED (alarmReceived) Host =
172.16.28.50, Community = ZmsTrapPrivate, Port = 162 Wed
Jan 30 16:47:18 EDT 2002: Forwarded
CARD_PROVISIONING_FAILED (alarmCleared) Host =
172.16.28.50, Community = ZmsTrapPrivate, Port = 162
```

The DZMS system generates the forwarded alarm log to the path: `install_directory/opt/weblogic/yyyy_mmm_dd_AlarmForward.log`.

Forwarded trap log

The DZMS system generates a forwarded trap log (*TrapForward.log*), listing information about traps that are forwarded to IP addresses on the network. The forwarded trap log lists entries sequentially by date and time, with the most recent trap appearing at the end of the file.

The following sample illustrates a trap log entry:

```
Fri Feb 01 09:13:10 EST 2002: sending trap to  
172.16.90.150
```

The DZMS system generates the forwarded trap log to the path: *install_directory/opt/weblogic/yyyy_mmm_dd_TrapForward.log*.

Server log

The DZMS system generates a server log (*zms.log*), listing server processes. For example, when the DZMS server is started, startup progress messages are written to the log. The log lists entries sequentially by date and time, with the most recent event appearing at the end of the file.

The DZMS system generates the server log to the path: *install_directory/opt/weblogic/zms.log*.

Trap log

The DZMS system generates a trap log (*trap.log*), listing all traps generated on the network. The trap log lists entries sequentially by date and time, with the most recent trap appearing at the end of the file.

These traps are defined in *zmsAlarm.mib*. [Table 18](#) lists the varbinds included in these traps.

The first five varbinds are standard to all DZS traps.

The last nine varbinds are defined for alarmReceived and alarmCleared traps defined in *zmsAlarm.mib* (faultServiceTraps).

Table 18: Trap information

Parameter	Description
SysUpTime	Elapsed time the device has been running when it generated the trap. This a standard SNMP varbind. Some SNMP tools may display this information in the trap header, rather than in the trap itself.
Trap OID	The trap OID. This a standard SNMP varbind. Some SNMP tools may display this information in the trap header, rather than in the trap itself.
Trap Version	Version number of the format of the trap.
Trap Sequence Number	Sequence number of the trap.
Trap Severity	Severity of the trap (critical, severe, moderate, or low).
Alarm Name	Name of the alarm.
Alarm Description	Description of the alarm.
Alarm Type	Type of the alarm.
Alarm Severity	Severity level of the alarm.
Alarm Timestamp	Time that the alarm was created.
Alarm Device	Name or IP address of the device that generated the alarm.
Alarm Shelf	Shelf that generated the alarm. A zero value indicates that the alarm was generated on a device.
Alarm Slot	Card slot that generated the alarm. A zero value indicates that the alarm was generated on a shelf.
Alarm Port	Physical port that generated the alarm. A zero value indicates that the alarm was generated on a card.

The following sample illustrates a trap log entry:

```
Tue Jan 08 15:07:12 EDT 2002: SNMP V2 Trap PDU
SNMP Version: Version 2C
Remote Host: 172.16.90.211
Remote Port: 41860
Community: ZmsTrapPrivate
Request ID: 1909962606
Timeout: 0
Retries: 0
Error Status: no error
```

SNMP PDU Variable Bindings:
Object ID: .iso.org.dod.internet.mgmt.mib-2.system.sysUpTime.0
TimeTicks: 0 hours, 0 minutes, 0 seconds.
Object ID:
.iso.org.dod.internet.snmpV2.snmpModules.snmpMIB.snmpMIBObjects.snmpTrap.snmpTrapOID.0
OBJID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneShelf.zhoneTrapShelfMonitorV2Traps.zhoneTrapCardStatusChange
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapVersion
INTEGER: version1 (1)
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapFlags
INTEGER: trapAcknowledge (0)
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapsTable.zhoneTrapsEntry.zhoneTrapsSequenceNumber
INTEGER: 1
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapsTable.zhoneTrapsEntry.zhoneTrapsSeverity
INTEGER: severe (2)
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapsTable.zhoneTrapsEntry.zhoneTrapsAllowedCategories
INTEGER: catAdmin (0)
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapShelf
INTEGER: 1
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapSlot
INTEGER: 3
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneSystem.zhoneTraps.zhoneTrapPort
INTEGER: 0
Object ID:
.iso.org.dod.internet.private.enterprises.zhone.zhoneGeneric.zhoneShelf.zhoneTrapCardStatus
INTEGER: cardRemoved (1)

The DZMS system generates the trap log to the path:
install_directory/opt/weblogic/yyyy_mmm_dd_trap.log.

DZMS error log

The DZMS system generates an error log (*ZMSErrors.log*), that stores any errors that were received due to improper provisioning (incorrect values) or device-related errors (such as SNMP time outs).

This log file is typically only useful for DZS development engineers.

The DZMS system generates the error log to the path: *install_directory/opt/weblogic/yyyy_mmm_dd_ZMSErrors.log*.

How DZMS MANAGES NETWORK ELEMENTS

This appendix explains how the DZMS managers and services function together to manage network elements. It describes the following managers and services:

- [Administration manager, page 103](#)
- [Configuration synchronization service, page 104](#)
- [Configuration manager, page 107](#)
- [Fault manager, page 108](#)
- [Monitoring service, page 116](#)

Administration manager

The function of the administration manager (ADS) is to enforce security through operator authentication and authorization.

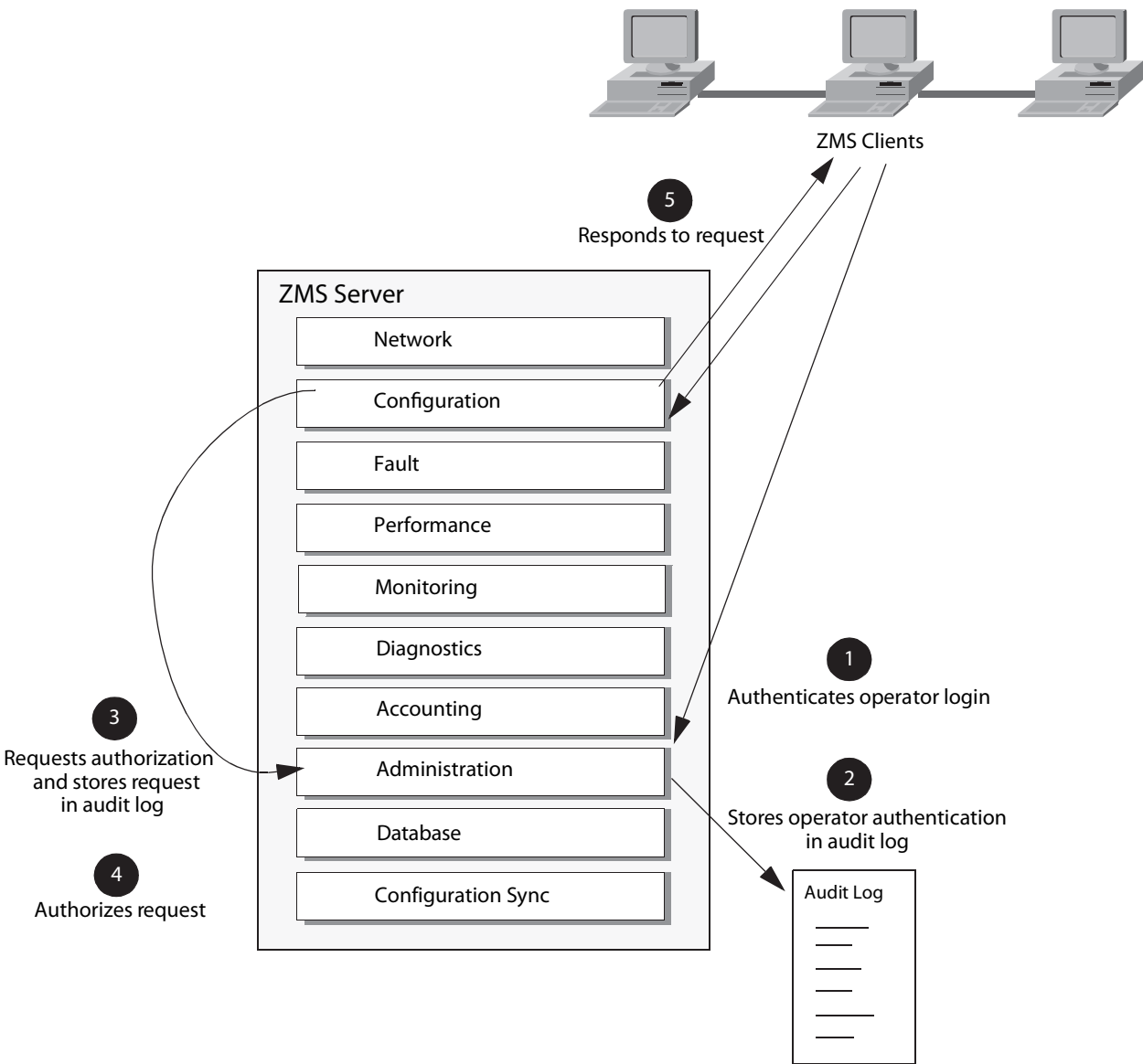
General process to managing operator requests:

- 1 When the operator logs into a DZMS client, ADS authenticates the operator.
- 2 ADS stores operator authentication requests in an audit log.
- 3 When a management request comes into DZMS from the operator, the appropriate manager (such as the configuration manager) requests ADS to authorize the request. The manager also stores the operator request in an audit log
- 4 ADS authorizes the request based on the operator's permissions.
- 5 The appropriate manager performs the request and forwards the response back to the client.

ADS is also responsible for managing operator accounts, including adding, modifying, activating, deactivating, and deleting accounts.

[Figure 2](#) illustrates how ADS authorizes and manages client requests.

Figure 2: Administering a management request



Configuration synchronization service

As resources are added to the network, the DZMS system uses the DZMS configuration synchronization service (CSS) to automatically synchronize elements in the system. The function of CSS is to ensure that the configuration information on a network device and in the DZMS database are synchronized.

When a DZMS object is first activated on the network:

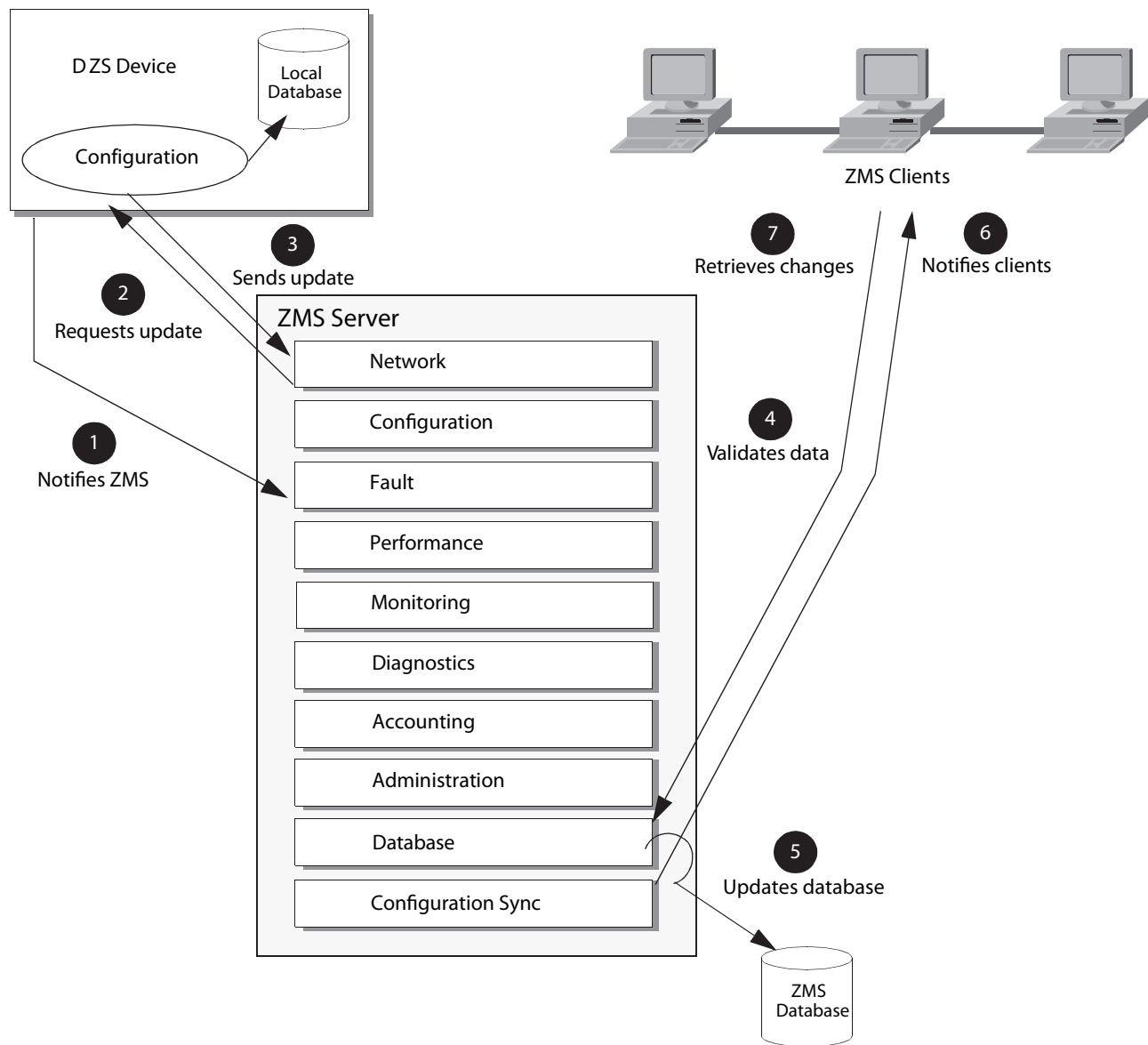
- 1 The device notifies CSS (by way of the fault service) that it is on-line.

- 2** CSS requests (by way of the network service) the device's configuration information.
- 3** In response, the device transfers the configuration information.
- 4** Once CSS receives the configuration information, it requests the configuration manager to validate the data.
- 5** Once the data is validated, CSS stores the data (by way of the database service) in the database.
- 6** CSS notifies each DZMS client that information has changed.
- 7** In response, the client retrieves the new information from the database service.

In the same way, anytime an object is modified at the device level (by way of the command line interface), the device notifies CSS in real time that a change has occurred. CSS requests the device's configuration information, and, once it receives it, notifies the database service and notifies DZMS client of the change.

[Figure 3](#) illustrates the synchronization process following a device-level change.

Figure 3: Synchronization following a device-level change



If the DZMS system is not active when an object is added or modified (such as during a network outage), the device stores the configuration changes in its local database. When the DZMS system comes back on-line, CSS notifies network devices that it is available. The devices send their configuration updates to CSS, which forwards it to the database service and to clients. The database service updates the DZMS database, and the clients update their displays with the information.

Configuration manager

The function of the configuration manager (CS) is to make configuration changes to network elements and provision subscribers and end-to-end services within the DZMS network.

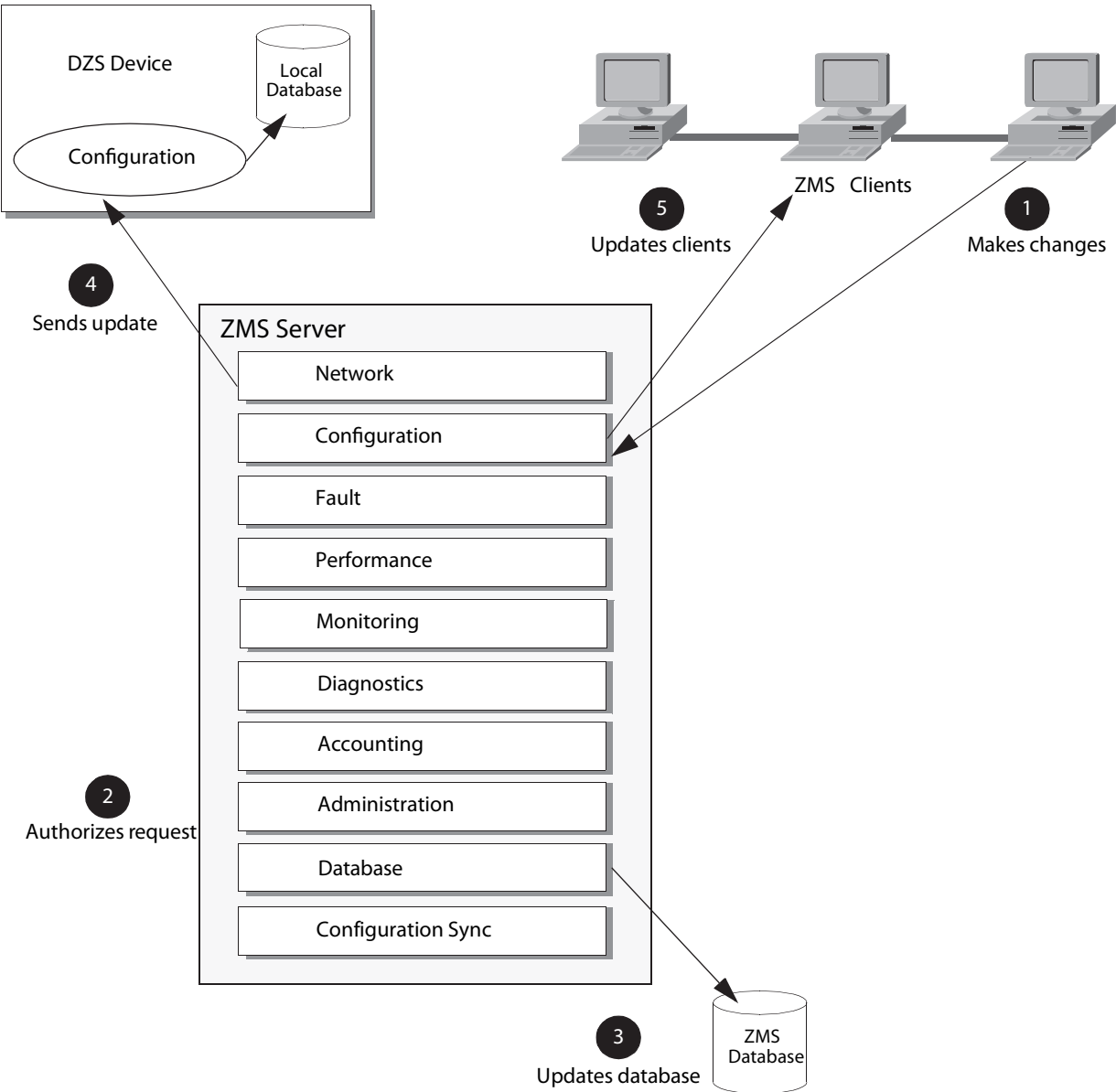
General process to provision resources:

- 1** NetHorizon makes a request to add, modify, or delete a network resource.
- 2** CS authorizes the configuration request (by way of the ADS).
- 3** CS performs the request and stores the data (by way of the database service) in the database.
- 4** CS sends the configuration update (by way of the network service) to the device.
- 5** CS notifies other ZMS clients of the change. Each operator's client displays updates to reflect the changes.

CS also handles element software downloads.

[Figure 4](#) illustrates how the CS provisions resources.

Figure 4: Configuring resources



Fault manager

The fault manager (FS) provides monitoring, logging, and notification of fault information (traps and alarms) on the DZMS network. FS also forwards traps to destination addresses, as needed.

A *trap* is an SNMP PDU containing real-time information about a predefined event occurring on a network device. A trap can indicate a problem such as a power supply failure or a performance problem. Or, a trap can indicate other dynamic information about network activity, such as a network object being brought down by a system administrator or a threshold level being exceeded. Traps exist for all objects, including devices, cards, physical ports, logical interfaces, permanent virtual circuits (PVCs), and so on.

The traps generated by DZMS objects are *reliable traps*. Each DZS device maintains a buffer of outgoing traps, so that it can resend any trap that is lost during transit.

An *event* is an SNMP PDU containing real-time information about a predefined event occurring in the management software. An event can indicate information about software processing, such as completion of card provisioning or completion of a partial update from the device.

A trap or an event can trigger an *alarm*, which is a human-readable message that notifies an operator or administrator of a network problem. A variety of alarm indications and statistics loggings are available for objects. For a list of traps that trigger alarms, see [DZMS Legacy Traps and Alarms on page 119](#).

Fault processing

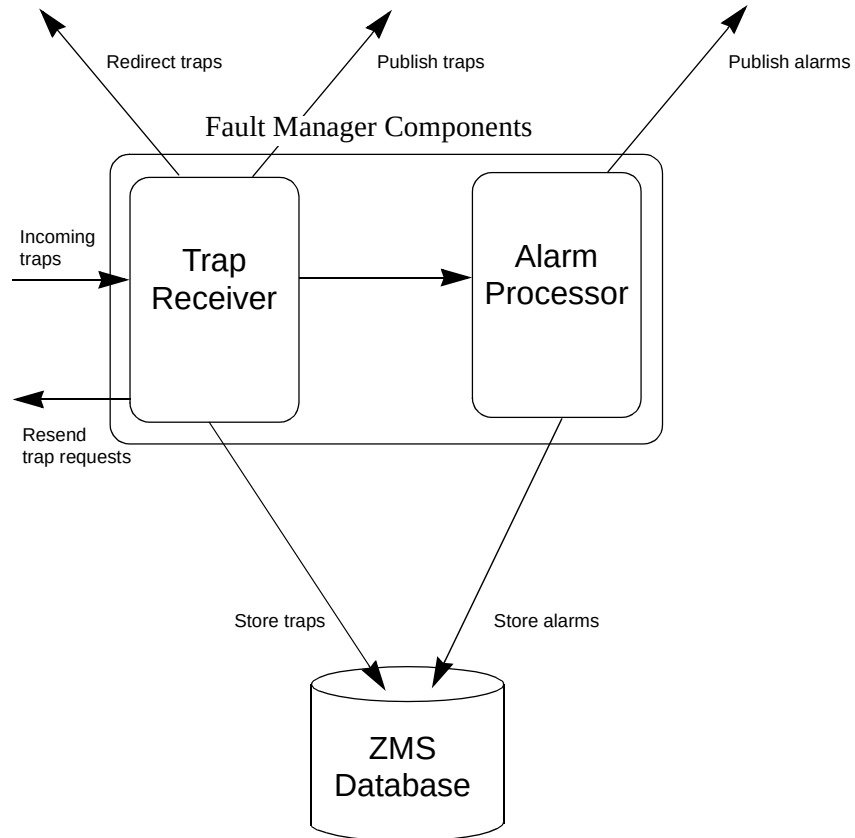
FS processes faults through a series of processors:

- Trap receiver, which collects and forwards traps.
- Alarm processor, which maps traps and events to alarms and applies faults to a set of rules to determine what action to take on each alarm.

DZMS allows the operator the flexibility of customizing the alarm processing and handling settings in the properties files (FSAlarmSeverity.properties, FSAlarmDescription.properties, and the FaultService property) on the DZMS server. See [Modify DZMS Alarm Descriptions on page 125](#).

[Figure 5](#) shows how FS components process faults.

Figure 5: Fault manager processors



Trap receiver

The trap receiver performs the following tasks:

- Processes traps
- Forwards traps

Trap processing

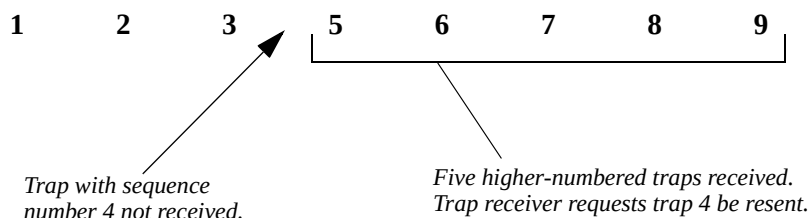
The trap receiver processes incoming traps in the order in which they arrive. The receiver uses the sequence number of each trap to verify that it has received all incoming traps. If the trap receiver detects that a trap is received out of sequence, the receiver reorders the traps into the correct sequence. If the trap receiver detects that a trap is missing, the receiver requests that the DZS device resend the trap.

The trap receiver uses the following criteria to determine if a trap is missing:

After five traps—If a trap with a particular sequence number is not received after receiving some number of higher-numbered traps (default: five).

For example, if the trap receiver receives traps in the order shown in [Figure 6](#), the receiver considers the trap with sequence number 4 missing when it receives the trap with sequence number 9 ($4 + 5 = 9$). The trap receiver requests the DZS device to resend trap 4.

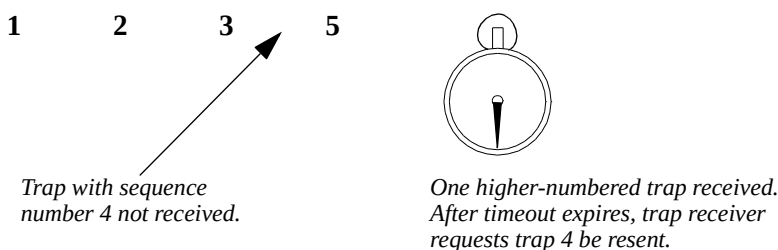
Figure 6: Handling reliable traps by sequence number



After timeout — If a trap with a particular sequence number is not received after receiving one higher-number trap and after a timeout (default: 30 seconds) elapses.

For example, if the trap receiver receives traps in the order shown in [Figure 7](#), the receiver considers the trap with sequence number 4 missing when it receives the trap with sequence number 5, and then the timeout expires (30 seconds). The trap receiver requests the DZS device to resend trap 4.

Figure 7: Handling reliable traps by timeout



The trap receiver stores information about each trap in a trap log and in the DZMS database, including the trap source (the object that generated the trap), the time at which the trap occurred, and the SNMP trap itself.

The trap receiver publishes each trap to DZMS clients.

The trap receiver forwards specific information about each trap to the alarm processor.

Trap forwarding

The trap receiver also forwards traps to other locations if any forwarding maps have been configured. As part of the process, the trap receiver adds the

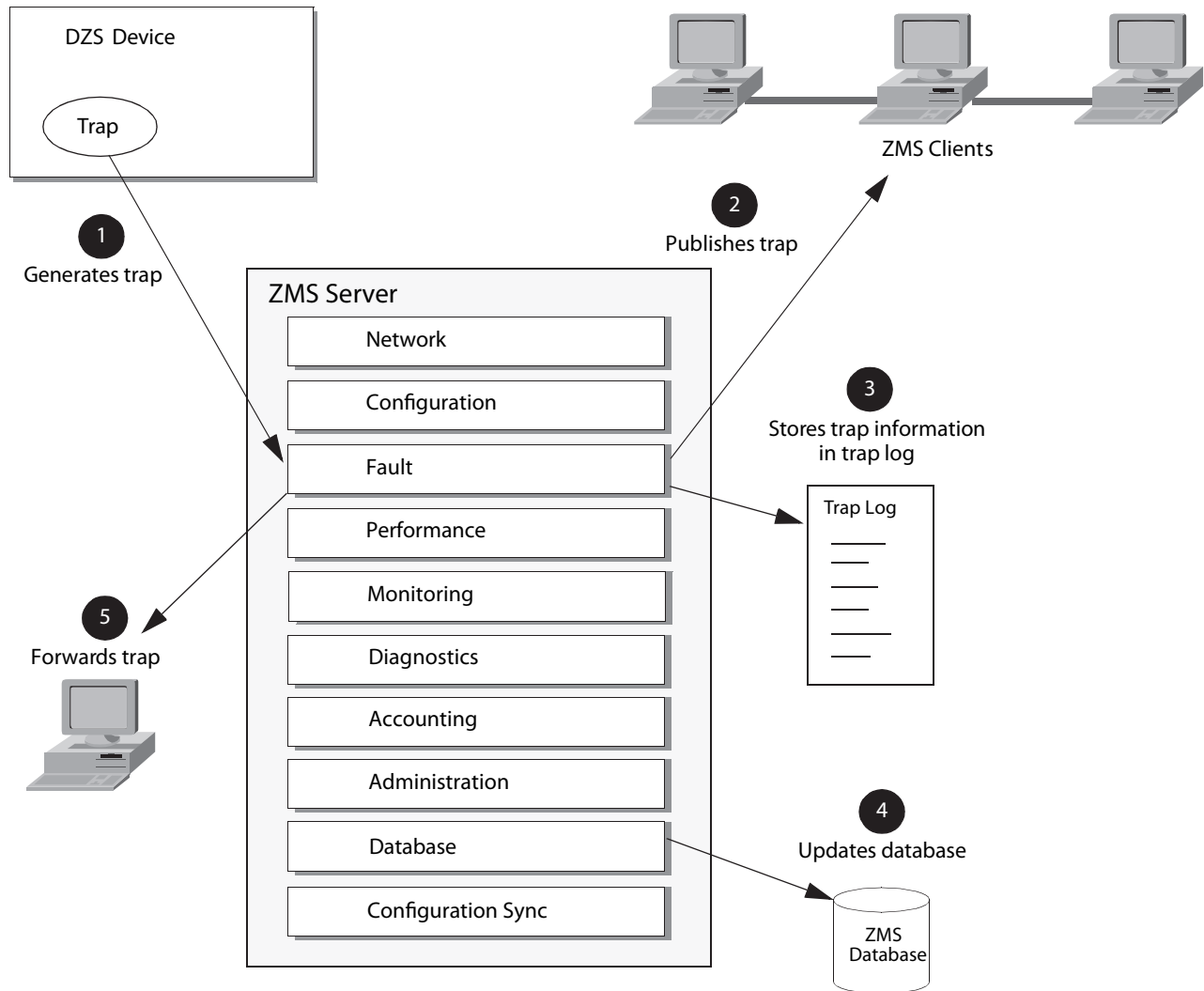
zhoneTrapsDestination variable binding to the trap to be forwarded. The variable binding specifies the IP address of the agent that generated the trap.

Environment variables in the *FaultService* property specify trap forwarding. Specific destination hosts are specified. Specific traps to be excluded from forwarding are also specified.

Figure 8 illustrates the general process that FS uses to process traps. The basic steps are:

General steps to process traps:

- 1** The device sends a trap that is received by the DZMS system (by way of FS).
- 2** The trap receiver publishes the trap to DZMS clients.
- 3** The trap receiver stores trap details in the trap log.
- 4** The trap receiver stores the trap (by way of the database service) to the database.
- 5** The trap receiver forwards the trap (including the zhoneTrapsDestination variable binding) to other registered workstations.

Figure 8: Processing traps

Alarm processor

The alarm processor performs the following tasks:

- Applies traps to a set of mappings that determine whether the trap becomes an alarm.
- Applies newly-created alarm to a set of rules that determine what action to take.
- Forwards alarms to the forward host manager, which sends a mail message to registered individuals, indicating that the alarm has occurred on the network (if configured to do so).
- Forward alarms to specific IP addresses (if configured to do so).

Alarm mapping

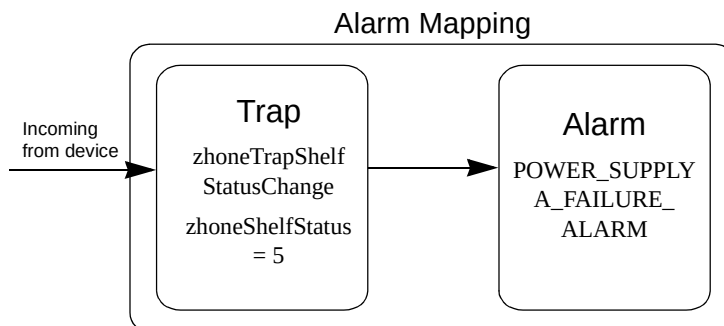
The alarm processor applies traps to a set of mappings that determine whether the trap becomes an alarm. Only a trap that meets the criteria specified by a particular map becomes an alarm.

An alarm mapping specifies:

- Name of the originating trap.
- Any restrictions for the values of the trap variables.
- Name of the generated alarm.

An alarm mapping may generate one or more alarms. For example, the trap `zhoneTrapShelfStatusChange` with trap variable `zhoneShelfStatus = 5` maps to the alarm `POWER_SUPPLYA_FAILURE_ALARM` (Power supply A failure).

Figure 9: Sample alarm mapping



For DS1 and Ds3 line status traps (`zhoneLineStatusChange` and `dsx3LineStatus` traps), if the value of the trap variable `zhoneLineStatus` is a power of 2 (such as, 1, 2, 4, 8, 16, and so on), the alarm processor generates a single alarm. If the trap variable value is a value other than a power of 2 (for example, 100), the alarm processor generates multiple alarms, one for each trap with `zhoneLineStatus` value within the variable value (in this case, 100). For example, the trap `zhoneLineStatusChange` with trap variable `zhoneLineStatus = 2` maps to the single alarm `LINE_RCV_FAR_END_LOF_ALARM`. The trap `zhoneLineStatusChange` with trap variable `zhoneLineStatus = 100` maps to three alarms (those generated for the variable values 4, 32, and 64). The sum of the values of `zhoneLineStatus` for these alarm-generating traps is 100.

Alarm severities

Alarm severities are defined in the `FSAlarmSeverity` property or by the trap that generates the alarm. Alarm severity levels can be changed by modifying the `FSAlarmSeverity` property. Alarm severities that are commented out in the `FSAlarmSeverity` property (with a “#” sign) derive their severity level from a trap. Alarm severities that are not commented in the `FSAlarmSeverity` property have the severity level defined in the file.

Applying rules

The alarm processor applies faults to a set of rules that determine what action to take. Applied rules may cause a trap to be cancelled or an alarm to be cleared.

For example, the alarm `POWER_SUPPLYA_FAILURE_ALARM` followed by the trap `zhoneTrapShelfStatusChange` with trap binding of `zhoneShelfStatus = 6` results in the action to clear the initial alarm.

An alarm can affect other alarms only while it is still active. Once an alarm is cleared, it can no longer clear other alarms.

Once the alarm has been applied to the rules, the alarm processor stores the alarm in an alarm log and in the DZMS database.

The alarm processor publishes each alarm to NetHorizon clients.

Alarm forwarding

The alarm processor also forwards alarms to specific IP addresses if any forwarding maps have been configured. The fault manager forwards alarm information wrapped in special traps to particular fault system(s) for processing.

The special traps defined for alarm forwarding are:

- `alarmReceived`, which is sent when a new alarm is generated in response to a trap from device.
- `alarmCleared`, which is sent when an existing alarm is cleared (either by a clearing trap or explicitly by an operator).

Environment variables in the *FaultService* property specify alarm forwarding, including destination hosts by alarm severity level and specific alarms to be excluded from forwarding. See [DZMS Legacy Traps and Alarms on page 119](#).

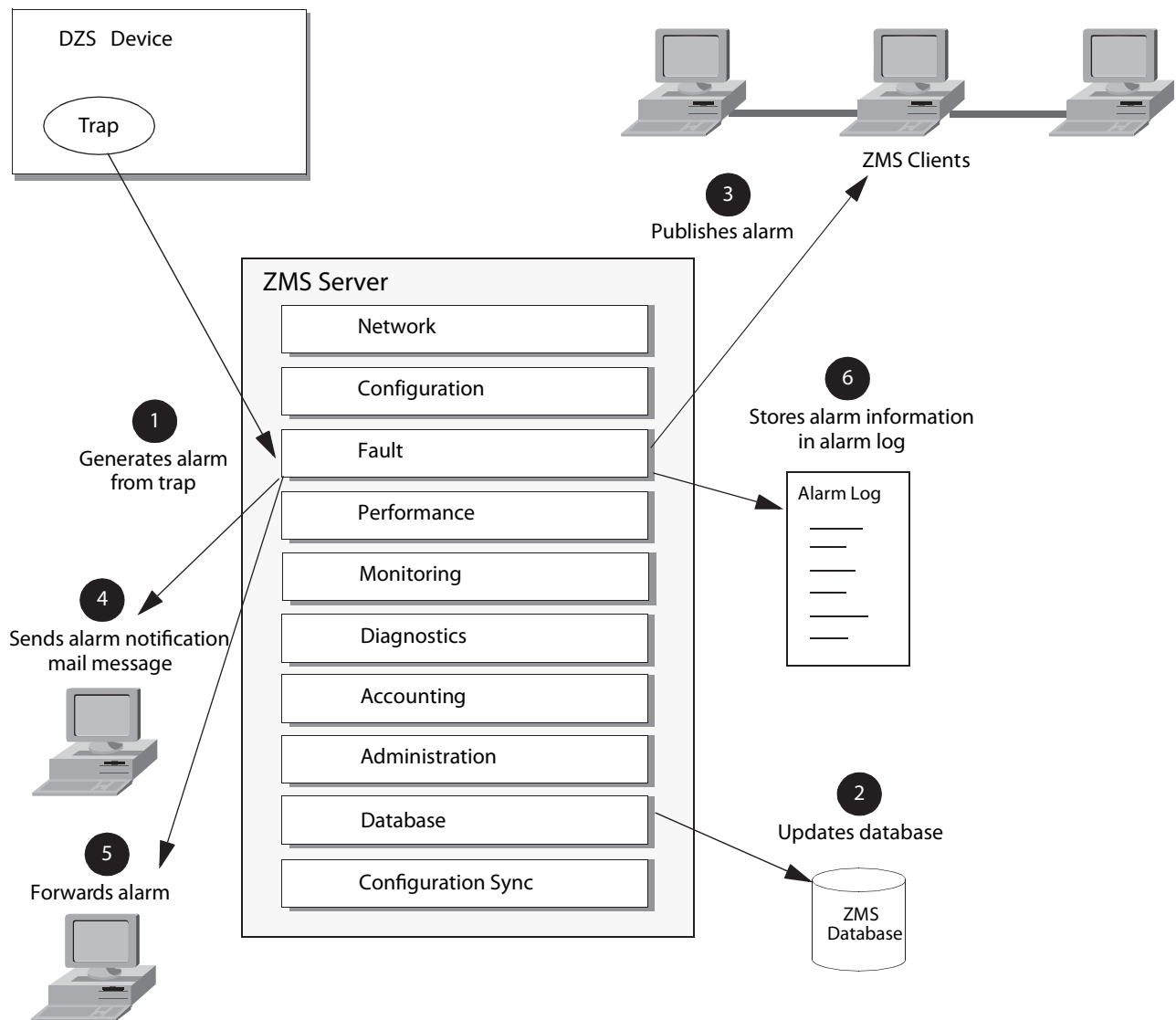
[Figure 10](#) illustrates the general process that FS uses to process alarms. The basic steps are:

General process to processing alarms:

- 1 Based on a trap received from a device, the alarm processor creates an alarm.
- 2 The alarm processor stores the alarm (by way of the database service) to the database.
- 3 The alarm processor publishes the alarm to DZMS clients.
- 4 If FS is configured to send alarm notification mail messages, the forward host processor forwards the alarm to the forward host manager, which sends a mail message to registered individuals, indicating that the alarm has occurred on the network.
- 5 If FS is configured to forward alarms, the alarm processor forwards the alarm wrapped in special traps to particular fault system(s) for processing.

6 The alarm processor stores alarm details in the alarm log.

Figure 10: Processing alarms



Monitoring service

The monitoring services (MS) monitors devices for changes in network connection status.

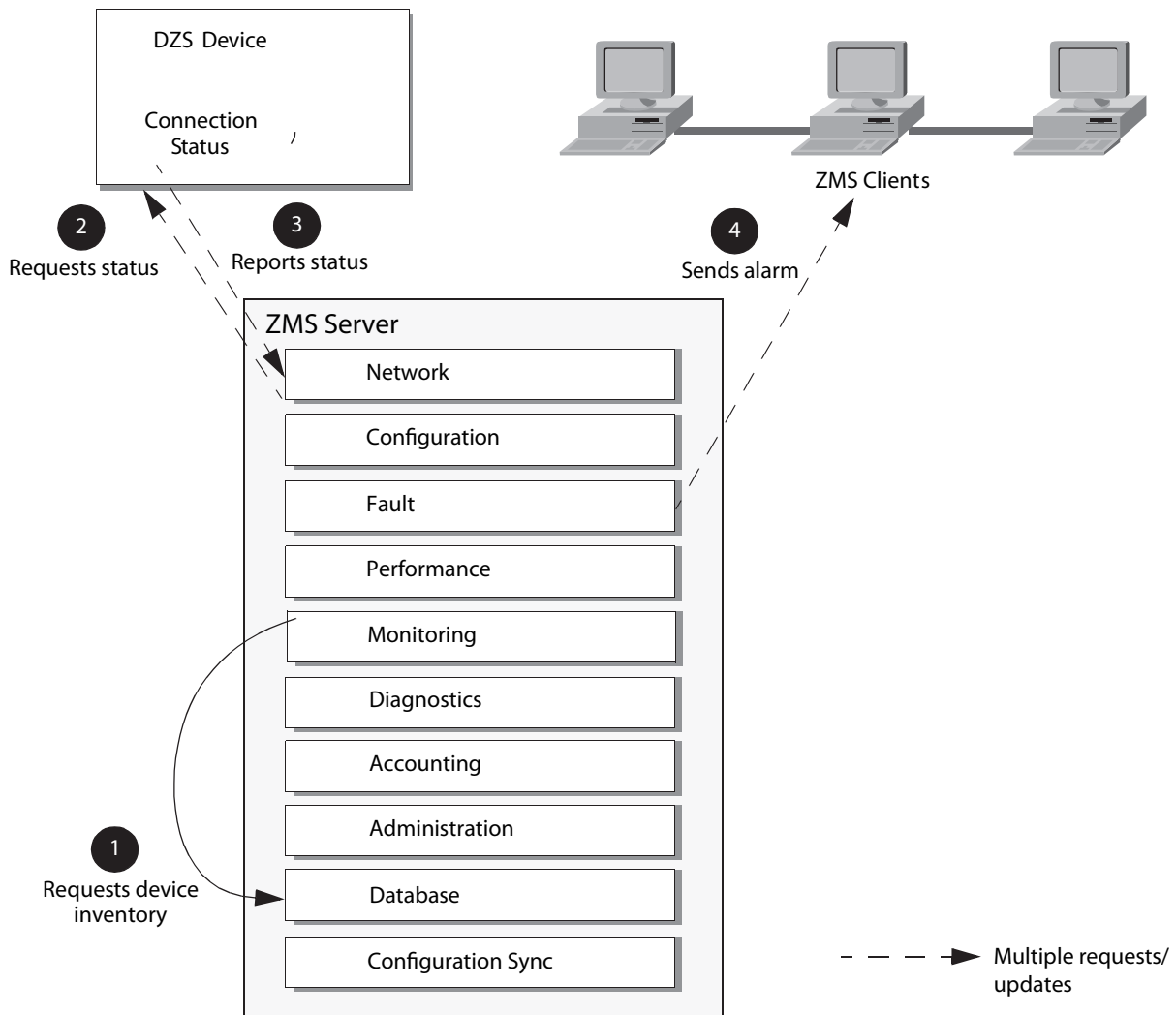
General process to monitor device status:

- 1** MS requests an inventory of devices from the database service.
- 2** Each polling interval, MS requests (by way of the network service) the network connection status from each device.

- 3 The device sends the connection status (by way of the network service) to MS.
- 4 MS sends an alarm (by way of the fault service) notifying DZMS clients of the change. Each operator's client display updates to reflect the trap.

Figure 11 illustrates how MS monitors for device network connection status.

Figure 11: Monitoring device network connection status





DZMS LEGACY TRAPS AND ALARMS

MXK changed the alarm method in 2.5 release. Prior to the change unless the alarm was listed in DZMS files it will not be displayed. After the change MXK provides all information so the new alarms do not need to be in the alarm files. If Severity or Description needs to be changed from defaults then the new alarm does need to be added to the alarmSeverity and/or the alarmDescription files.

This appendix describes the DZMS legacy alarms only.

- [DZMS alarms, page 119](#)
- [ZNID 4200 alarms {{it is for EZtouch, remove it?}}, page 122](#)
- [ZRG alarms {{product no longer support. remove it?}}, page 122](#)
- [DZMS server disk space alarms, page 123](#)

DZMS alarms

DZMS cannot reach device

Alarm	Trap	Severity
The device is not reachable	N/A	Critical

DZMS login failure alarm

Enabling the option **Generate ZMS Alarm on Login Failure** in the **Modify Security Policy Configuration** dialog box by admin user is required for DZMS login failure alarm raising. Every time user fails on the login, a DZMS login failure alarm will be raised, and an alarm notification email with username and host IP information will be sent out if alarmMailServer, alarmMailSender, and zmsAlarmMailRecepients are specified in the property file.



Note: One DZMS login failure alarm for each username.



Note: Multiple login failures will be displayed in the Description column of the **Alarm Viewer** window with the number of login attempts.

For example, **Reoccurred(2): Client user login failure**.

DZMS can now retrieve the DZMS login failure alarm in this release:

Alarm	Trap	Severity
Client user login failure	N/A	Warning

DZMS config sync alarms

A full config sync update has failed

An object in the device configuration database has changed by CLI modification. A CLI modification activates a partial config sync.

DZMS possible splitter fiber cut alarm (Optional)

The Possible Splitter Fiber Cut alarm is supported for both NetHorizon and Web Client and optional.

MXK generates the passes the ONU Link Down alarms to DZMS. By adding the Possible Splitter Fiber Cut alarm, if all ONUs attached under the same splitter showed Link Down alarms then DZMS will raise the alarm stating, for example:

Splitter 02 on OLT 1-10-1-0 has a possible fiber cut

If any of these ONUs should recover and report to DZMS with ONU up notifications, then the Possible Splitter Fiber Cut alarm will be automatically cleared.

The Possible Splitter Fiber Cut alarm can be used in the multi-splitter environment or the single-splitter environment.

By default, the Possible Splitter Fiber Cut alarm is disabled. After enabling it, users can specify the Splitter ID and ONU position values. Based on the Splitter ID value, DZMS knows how many and which ONUs are attached with which splitter.

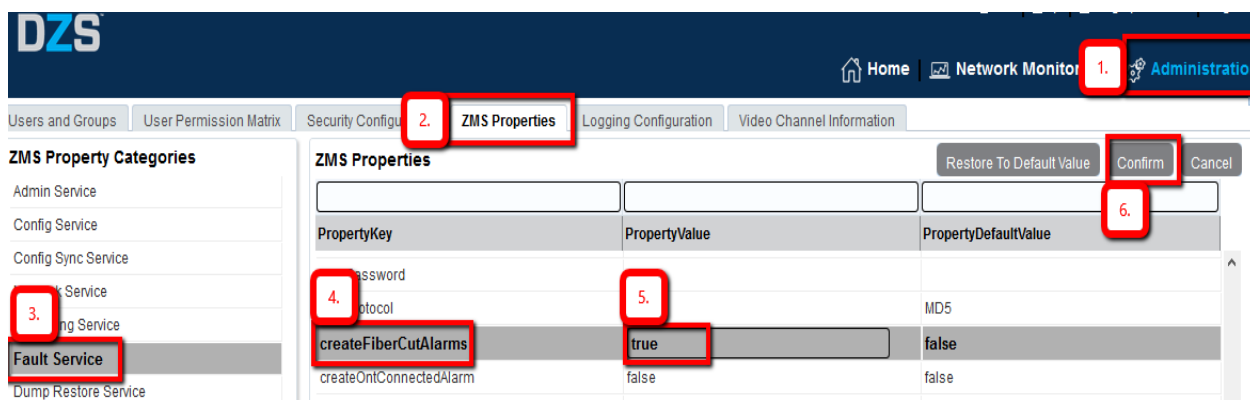
Configuring the Possible Splitter Fiber Cut Alarm

- 1 Enable the Possible Splitter Fiber Cut Alarm as Admin.

The Possible Splitter Fiber Cut alarm is optional and can be controlled using the DZMS parameter *createFiberCutAlarms*, by default the parameter is false and the alarm is disabled.

To enable the alarm, must execute the following procedure as Admin:

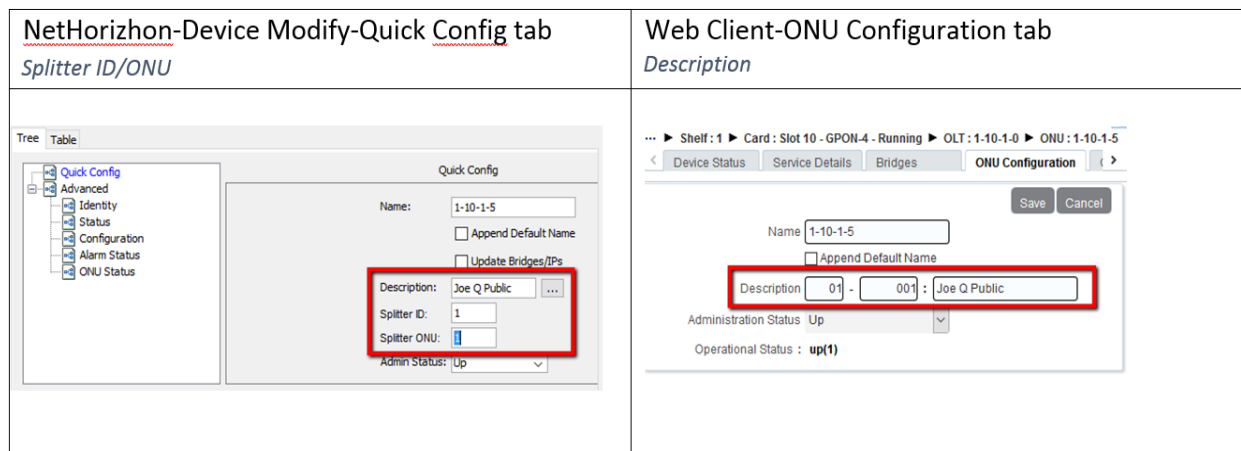
- a Choose Administration from top navigation menu.
- b Choose Fault Service from DZMS Property Categories.
- c Scroll to createFiberCutAlarms PropertyKey.
- d Edit PropertyValue to true.
- e Click Confirm to save change.
- f A general warning appears and shows that DZMS needs to be restarted. Although this is good practice when changing DZMS Properties, in this case you can ignore the warning. Fiber Cut Alarming is now active.



2 Add the Splitter information to ONU description.

After enabling the Fiber Splitter Cut alarm, the ONU description editing will show extra text boxes to allow entry of Splitter ID and ONU position values.

Note that there is no DZMS control over Splitter ID or ONU position values. These values are defined by users. Since only the Splitter ID value is used for alarm purposes, the ONU position is provided to allow the user to use as place holder for outside plant information as to where the ONU is attached at the splitter in the field.



- 3 Splitter ID and ONU position values are displayed in the tree view for ONU.

NetHorizon	Web Client
- Slot 10 - GPON-4 - Running - OLTs - OLT 1 : 1-10-1-0 (1 unassigned SNs) - ONUs - ONU 1 : 1-10-1-1 - ONU 2 : 1-10-1-2 (SN:ZNTS03380DD5)(ME:zhone-2426) - ONU 3 : 1-10-1-3 (01-003:117 W Main St - 1st State Bank) - ONU 4 : 1-10-1-4 (01-002:Jane Doe) - ONU 5 : 1-10-1-5 (SN:ZNTS03380C7B)(ME:zhone-2426) (01-001:Joe Q Public) - ONU 6 : 1-10-1-6 - ONU 7 : 1-10-1-7	- Un-ProvisionedONUs 1-10-1-1 1-10-1-2(SN:03380DD5) (Model:2426A) 1-10-1-3 (01-003:117 W Main St - 1st State Bank) 1-10-1-4 (01-002:Jane Doe) 1-10-1-5(SN:03380C7B) (Model:2426A) (01-001:Joe Q Public) 1-10-1-6

ZNID 4200 alarms {{it is for EZtouch, remove it?}}

DZMS supports the following ZNID 4200 alarms.

Alarm	Trap	Severity
The ZNID is receiving power from commercial power. This is the normal mode of operation.	znidBatteryRelayNotification znidBatteryRelayStatus = 1	Minor
The ZNID is receiving power from battery.	znidBatteryRelayNotification znidBatteryRelayStatus = 2	Minor
The battery has been reduced to the point that roughly 20% of the available runtime is available.	znidBatteryRelayNotification znidBatteryRelayStatus = 4	Minor
The battery has failed its periodic test. The battery should be replaced as system availability has been compromised.	znidBatteryRelayNotification znidBatteryRelayStatus = 6	Minor
The battery is disconnected. The ZNID will not be supported if the commercial power fails.	znidBatteryRelayNotification znidBatteryRelayStatus = 8	Minor
The UPS unit is disconnected. The ZNID will not be supported if the commercial power fails.	znidBatteryRelayNotification znidBatteryRelayStatus = 16	Minor

ZRG alarms {{product no longer support. remove it?}}

Alarm	Trap	Severity
The ZRG is receiving power from commercial power. This is the normal mode of operation.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 1	Minor
The ZRG is receiving power from battery.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 2	Minor

Alarm	Trap	Severity
The battery has been reduced to the point that roughly 20% of the available runtime is available.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 4	Minor
The battery has failed its periodic test. The battery should be replaced as system availability has been compromised.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 6	Minor
The battery is disconnected. The ZRG will not be supported if the commercial power fails.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 8	Minor
The UPS unit is disconnected. The ZRG will not be supported if the commercial power fails.	zrgBatteryRelayNotification zrgBatteryRelayStatus = 16	Minor

DZMS server disk space alarms

This feature is for fetching DZMS disk space alarms.

There are 4 DZMS properties in DZMS Properties / Fault Service called:

- `fetchZmsDiskSpaceInterval`
The interval in minute for fetching DZMS disk space. Setting this property to "0" will disable fetching disk space. (default is 10s)
- `zmsDiskSpaceMinorThreshold`
The minor "usableSpace/totalSpace" percentage ratio threshold. (default is 10%)
- `zmsDiskSpaceMajorThreshold`
`zmsDiskSpaceMajorThreshold`: The minor "usableSpace/totalSpace" percentage ratio threshold. (default is 20%)
- `zmsDiskSpaceCriticalThreshold`
`zmsDiskSpaceCriticalThreshold`: The critical "usableSpace/totalSpace" percentage ratio threshold. (default is 30%)

Like changing all other DZMS properties user needs to log in as "admin" user and then modify this properties.

The alarms are shown in NH's View / Faults / View Alarms panel.



MODIFYING DZMS ALARM SEVERITIES AND DESCRIPTIONS

This appendix describes the following two topics:

- [Modify DZMS Alarm Descriptions on page 125](#)
- [Modify DZMS alarm severity levels on page 126](#)

Modify DZMS Alarm Descriptions

You can customize individual alarm descriptions to include one or more attribute values by modifying the `/zms/lib/com/zhone/zms/properties/FSAlarmDescription.properties` file.

Adding an attribute to DZMS alarm descriptions

To add an attribute to an alarm description, use the format `$$attribute_name$$`, where `attribute_name` is an existing attribute on the object generating the alarm. For example, to identify what particular object generated an alarm, you can add the object name to the alarm description:

The status of AAL type 2 PVC (`$$name$$`) (`vpi=$$vpi$$/vci=$$vci$$`) has gone down

As shown in the above example, the text on the left of the "=" is the DZMS alarm name that is used to uniquely identify the alarm. The text on the right side of the "=" is the alarm text. This alarm text contains two `$$attribute_name$$`: `$$vpi$$` and `$$vci$$`. The value for that attribute will be added into the text.

you would see:

The status of AAL2 PVC pvc_la vpi=0/vci=100 has gone down

For a list of attribute names contact the DZS Quality & Service.

To add an attribute to DZMS alarm's text:

- 1 Search the `FSAlarmDescription.properties` file for the line containing the current alarm text.
- 2 The text on the right side of the "=" is the alarm text. You can change the alarm text as you desired.

For example, add a new attribute `vci=$$vci$$` to the alarm text:

Enter: **The status of AAL type 2 PVC (\$\$name\$\$)
(\$\$vpi=\$\$vpi\$\$/\$\$vci=\$\$vci\$\$) has gone down**

Modify DZMS alarm severity levels

The value of the severity level that DZMS assigns to each alarm can be modified by updating `/zms/lib/com/zhone/zms/properties/FSAlarmSeverity.properties` file.

Table 19: DZMS alarm severity level values

Alarm severity levels	Values
Critical	1
Major	2
Minor	3
Warning	4
Informational	5

Modifying DZMS alarm severity levels

To modify the severity level of a DZMS alarm, use the following procedure:

- 1 Obtain the alarm name by searching `FSAlarmDescription.properties` for the current alarm text.
The text on the left side of "=" is the DZMS alarm name.
- 2 Search `FSAlarmSeverity.properties` for the DZMS alarm name.
- 3 If it exists, replace the current alarm severity value with the desired value.

For example:

Enter: **SHELF_FAN_A_FAILURE_ALARM=2**

It will change a "FAN A failure" alarm to be a major alarm.

- 4 If it does not exist, add the following line into the file, then continue to Step 3:

ZmsAlarmName=severityValue

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